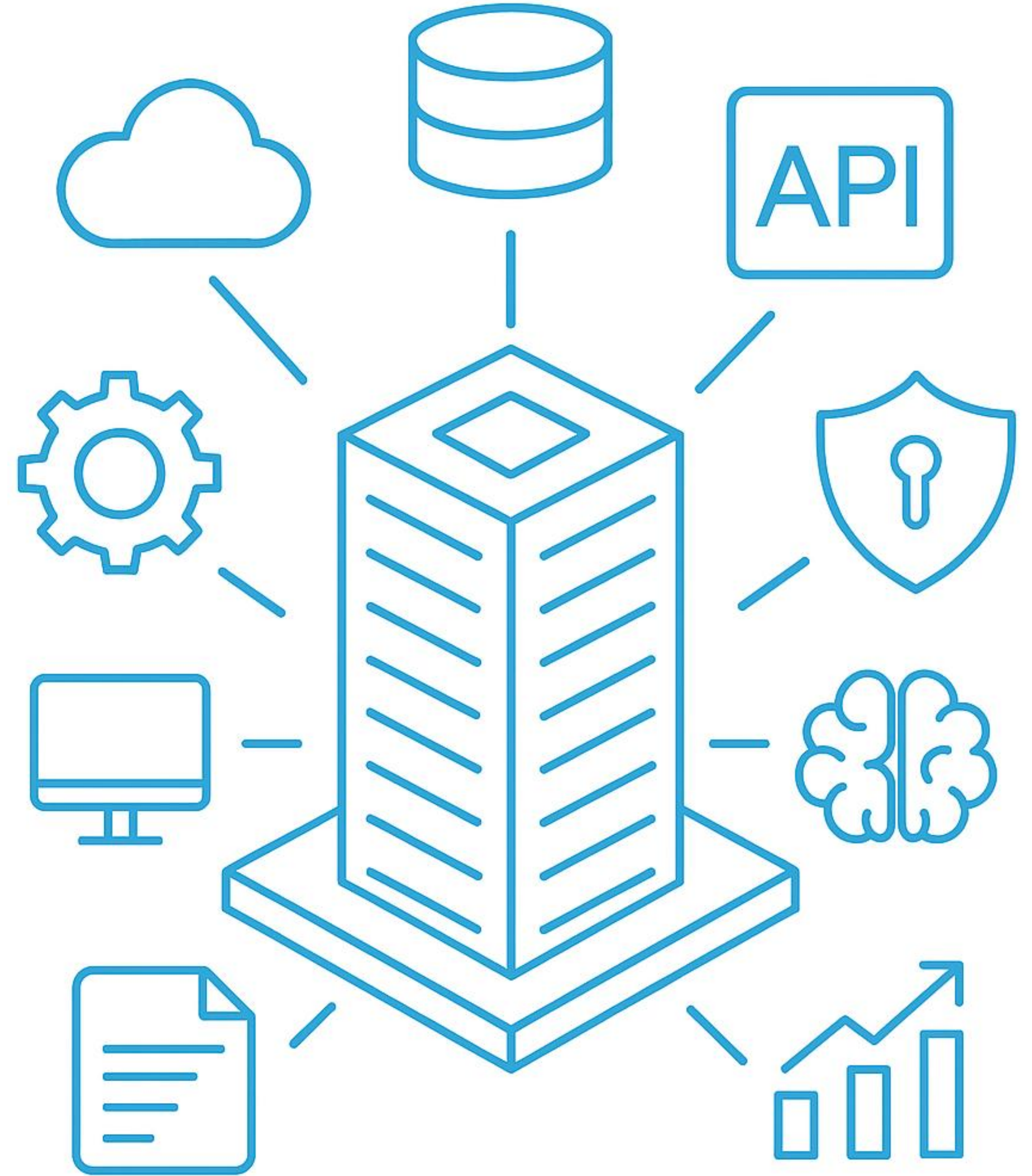
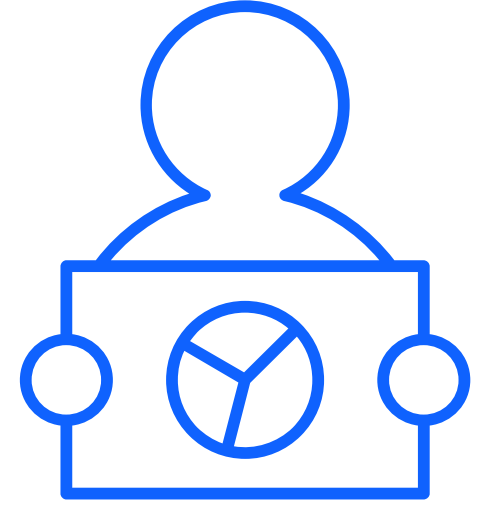
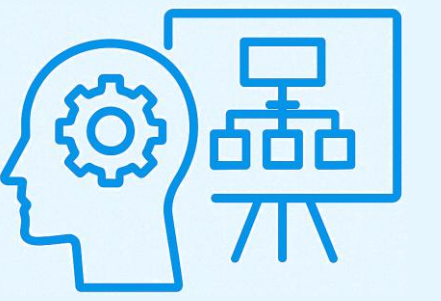


Operational Aspect

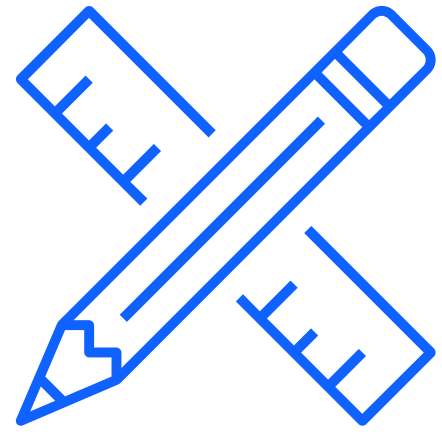
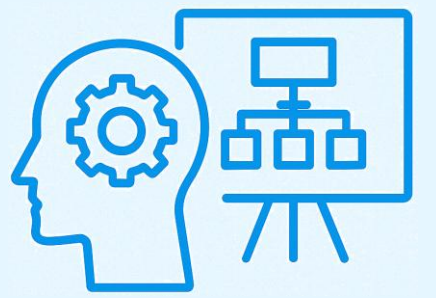
Version 7





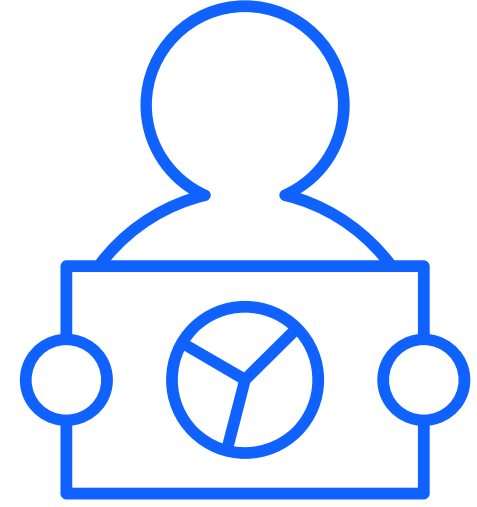
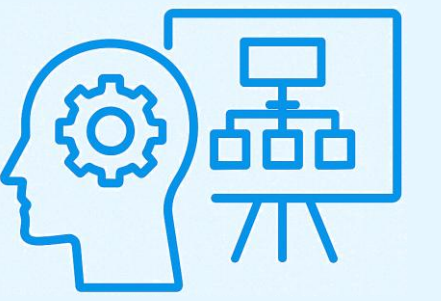
→ **Introduction and objectives**

- ↘ What is Operational Modeling (OM) and why do we need it?
- ↘ How does OM evolve and how is it represented?
- ↘ Constructing an Operational Model
- ↘ Summary and references



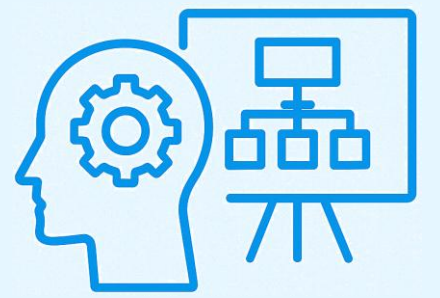
At the end of this module, you should be able to:

- Understand and explain the:
 - Purpose of the operational aspect of architecture
 - Key concepts and artifacts of the operational aspect
 - Relationships between the functional aspect and the operational aspect
 - Techniques for creating Operational Models from organized sets of requirements
- Assist an experienced Architect in constructing and documenting the Operational Model
- Review an existing Operational Model document for completeness



- Introduction and objectives
- ➔ **What is Operational Modeling (OM) and why do we need it?**
- How does OM evolve and how is it represented?
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- Summary and references

The Operational aspect is one of the two complementary aspects of architecture



• The Functional aspect of Architecture <- Previous

Module

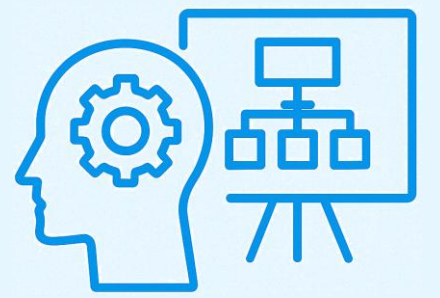
- Describe the structure of components (of all kinds) related to actual functionality,
- ...their dynamic behavior (collaboration),
- ...and their interfaces.
- Embodied in the Component Model work product



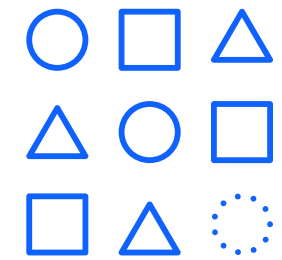
• The Operational aspect of Architecture <- This Module

- Describes what runs where (component placement).
- ...and includes the network topology (hardware nodes, locations, etc.),
- It ensures the achievement of the solution's service level characteristics (performance, availability),
- ...and describes the management and operation of the IT system.
- Embodied in the Operational Model work product

Functional systems must be deployed operationally - either using new or an existing infrastructure/cloud services

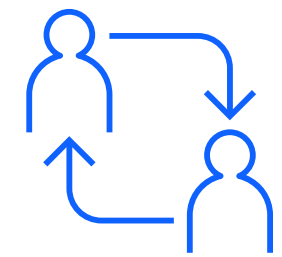


The design of a system's operational aspect may be challenging to develop or understand.



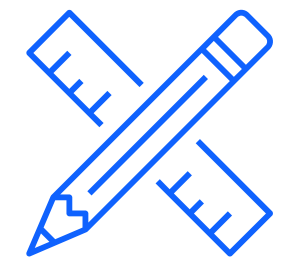
Many competing concerns must be resolved:

- NFR's: for example, performance, availability, manageability, security
- Constraints: for example, affordability, standards compliance, existing Infrastructure

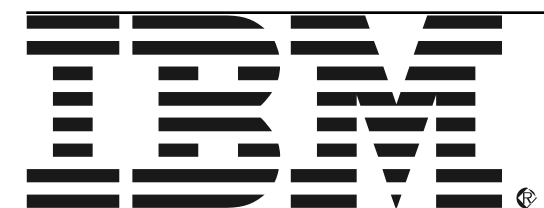
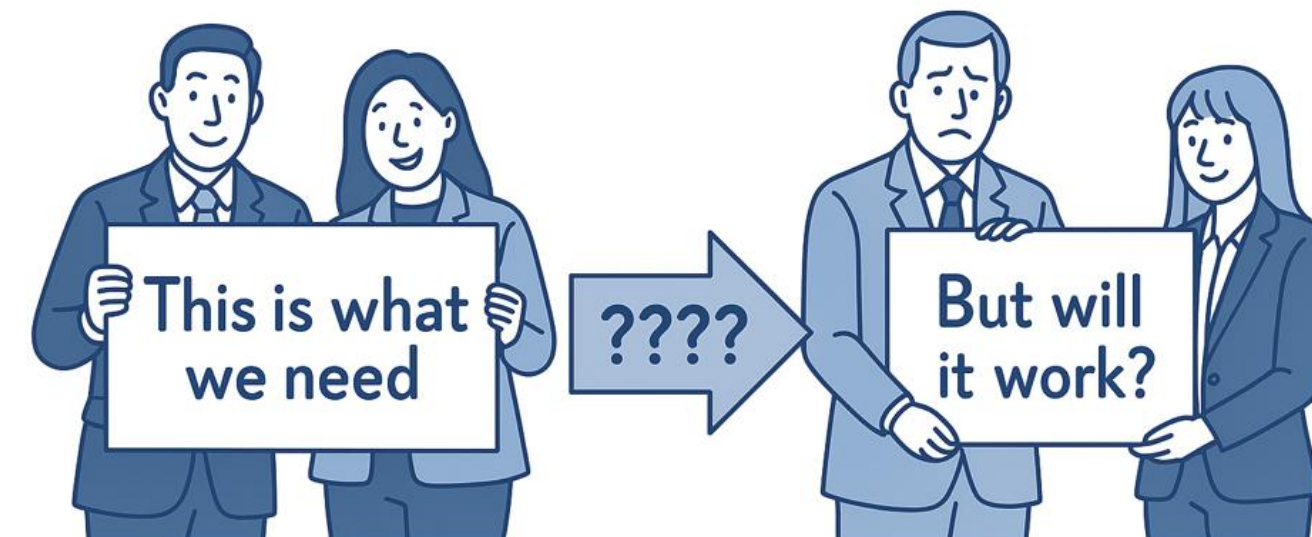


Multiple activities must be coordinated:

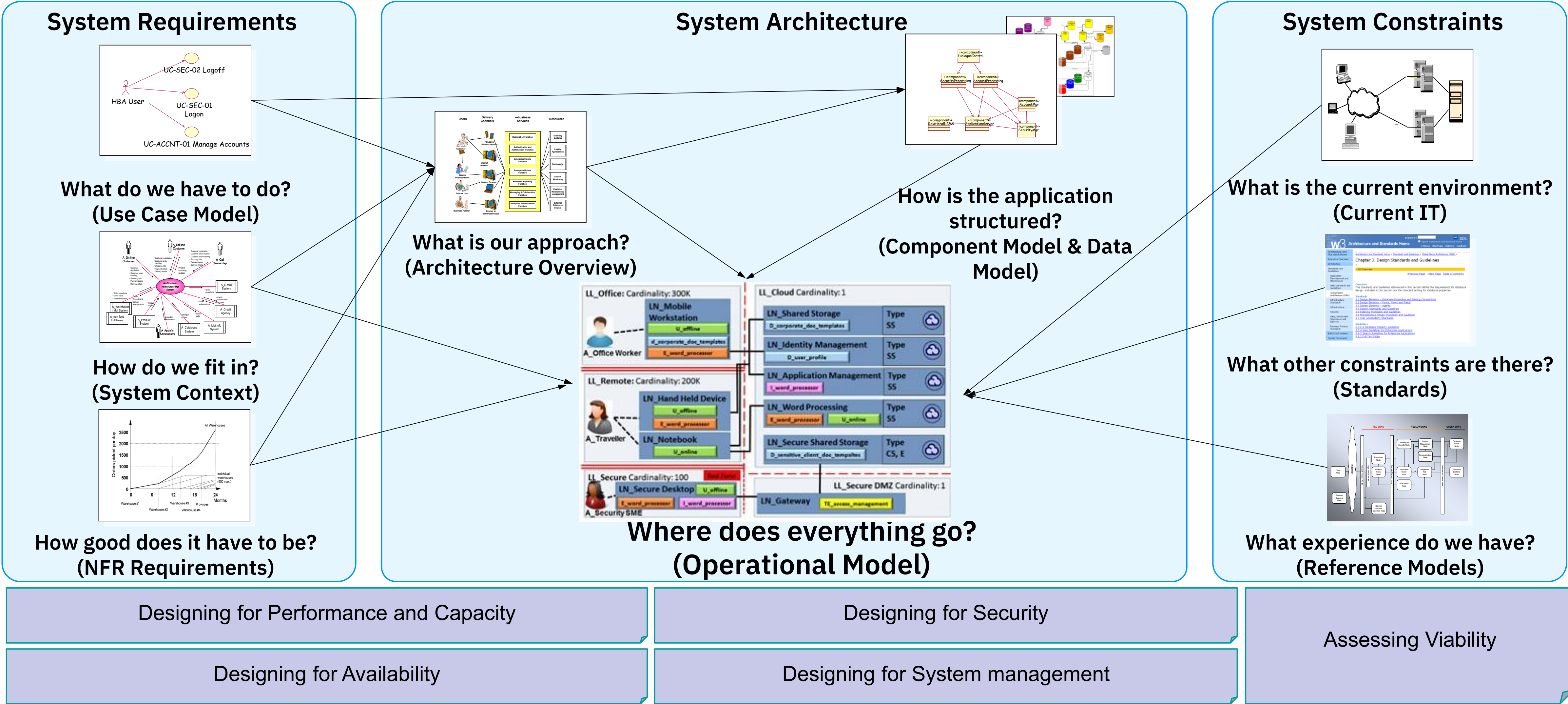
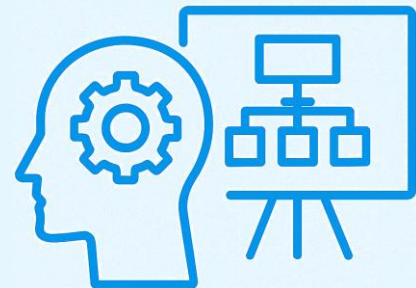
- **Detailed design:** capacity planning, software and data distribution, network, hardware, service management
- **Commercial:** planning, pricing, procurement, service level agreement

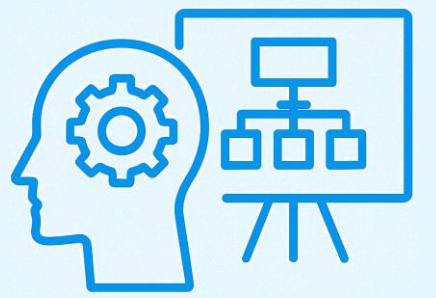


The design of a system's operational aspect may be challenging to develop or understand.

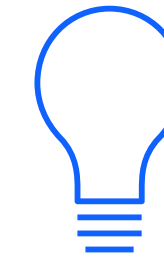


Operational Modeling helps ensure the system's non-functional (NFR) requirements are delivered, within all constraints



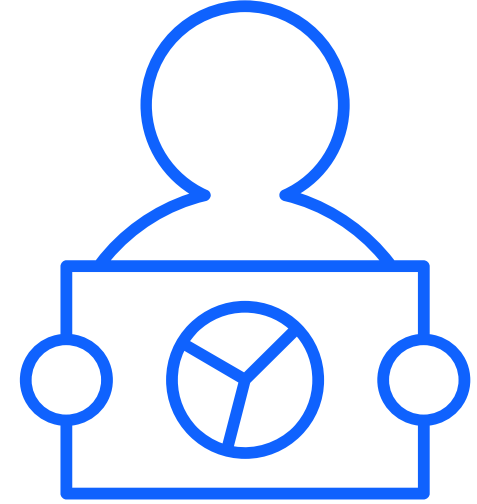
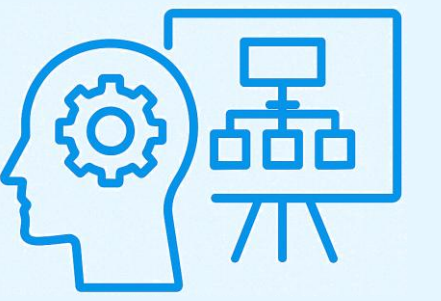


Time for a question



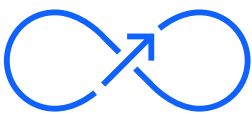
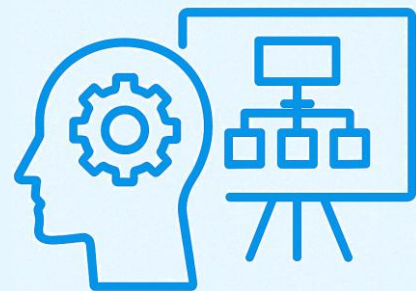
Select the correct statement from the following:

- A. The Operational Model focuses on the structure of the solution by identifying components and their collaborations.
- B. Service level requirements are delivered through the deployment of the components identified in the Operational Model.
- C. The Operational Model describes how components are supported to meet the service level requirements.
- D. The Operational Model is used as input when building the Location Model and the Node Model.



- Introduction and objectives
- What is Operational Modeling (OM) and why do we need it?
- ➔ **How does OM evolve and how is it represented?**
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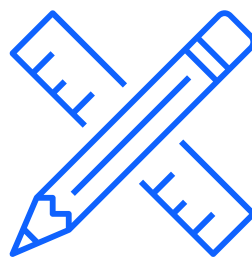
The Operational Modelling technique consists of multiple steps



1. Understand the non-functional requirements

What NFRs need to be met?

1M transactions per hour



2. Conceptualize from an application perspective (ALOM):

Identify application level nodes and their placement.

Note: Application Logical Operational Model (ALOM) is not discussed in this lecture – see the Operational Modelling class for details on ALOM.

Finance

Warehouse Management

Customer Relationship

Application



3. Conceptualize the Logical Operational Model (LOM):

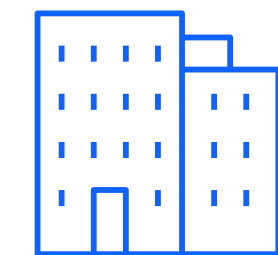
- Determine where all of the logical application components are deployed.
- Determine what logical technical components are required and where they will be deployed to support the logical application components

ESB

Load balancer

Database

Logical



4. Transform into the Prescribed Operational Model (POM):

Determine real technologies to deliver all of these specifications.

SAP

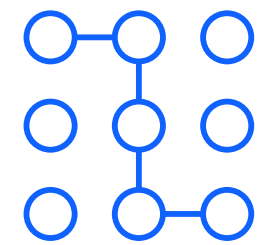
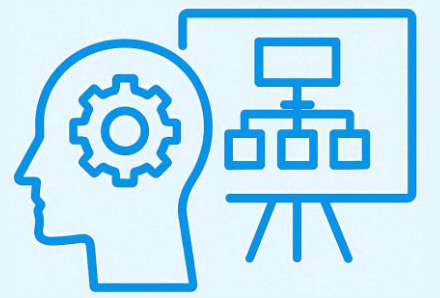
Message Broker

F5 BIG-IP LTM-4200v

DB2

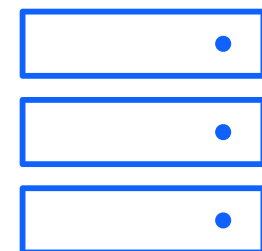
Prescribed

To achieve this goal, we need a structured, formal language to describe the elements featured on the Operational Model



The Operational Model represents the system’s “infrastructure architecture” using a variety of model elements, including:

- The geographic structure of the [locations](#) and their [borders](#), over which the IT system will be deployed and operated
- The placement of the system’s [nodes](#) into these locations
- The deployment of the system’s components across these nodes, using [deployment units](#)
- The [connections](#) between nodes
- The organization of the system’s elements into [zones](#)

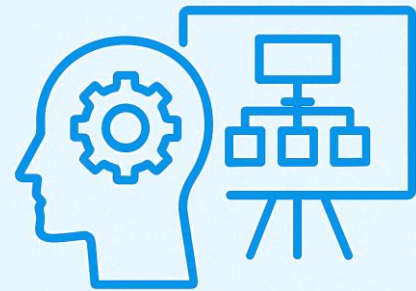


In addition to this description of the deployed system, the OM also documents:

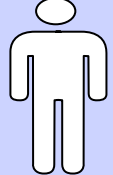
- The overall [prescribed configuration](#) of the technologies and products necessary to deliver the functional and nonfunctional requirements of the IT system
- Sizing and other [hardware specifications](#) for all the computers, storage devices, and network technologies



Operational Modeling (OM) notation used throughout this lecture and upcoming exercise



Actors & Infrastructure

Type	Logical Example	Prescribed Example
Location	LL_Office	PL_IBM Bolumbia Head Office
Node	LN_Mobile Workstation	PN_Apple Laptop
Human Actor	A_Office Worker  	
System Actor	 A_Respondent PC 	

Connections

No permitted connection

External intermittent connection

Internal high speed connection

External high speed connection

Cloud

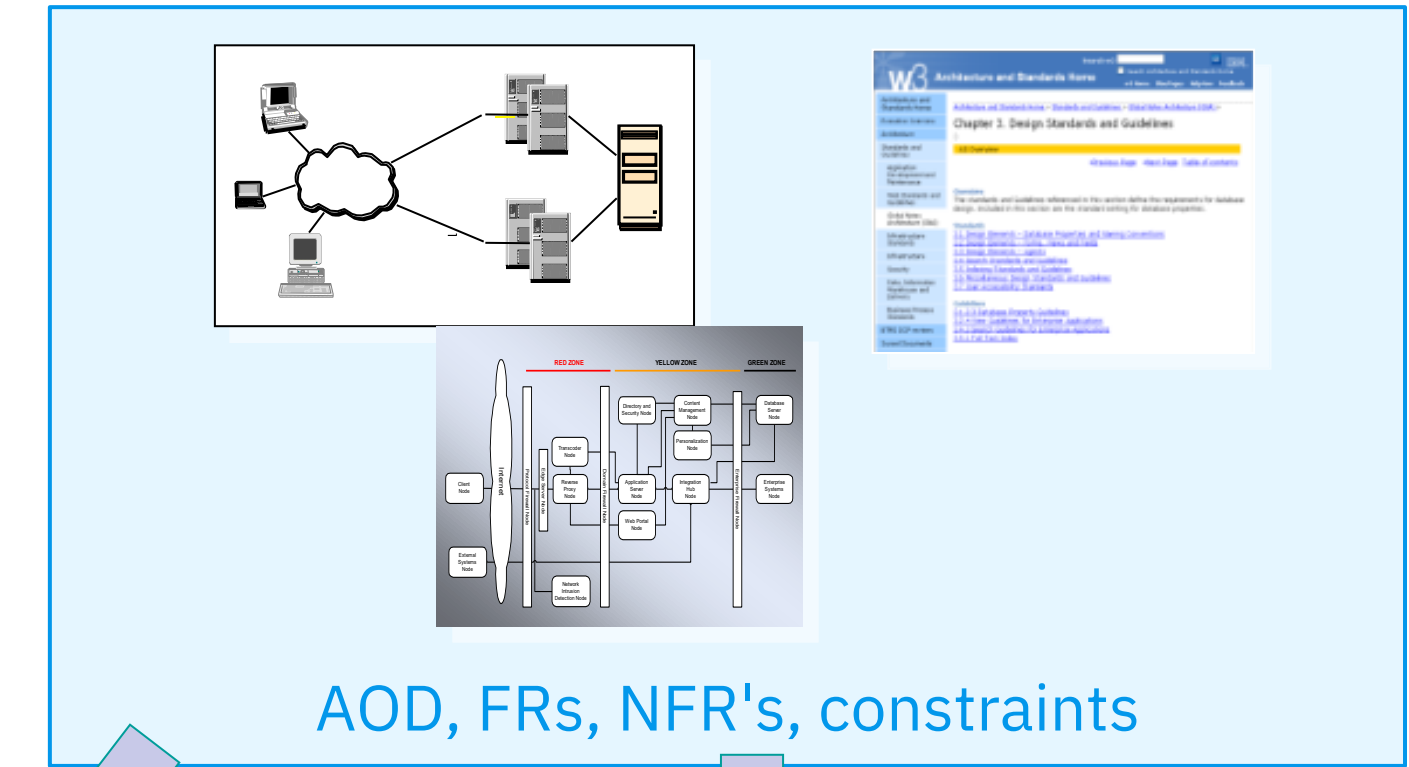
Type	Example
Offering	 On-premises  IaaS (Infrastructure)  SaaSH Hosted
	 SaaSM (Software) Managed  External/Web Resource
Support	SS - Standard Service Level Agreement(s) CS – Custom Service Level Agreement(s)
Data	E – Encrypted, NE – Non-Encrypted, AU - Audited, AR – Archiving Required

Deployment Units

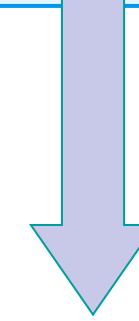
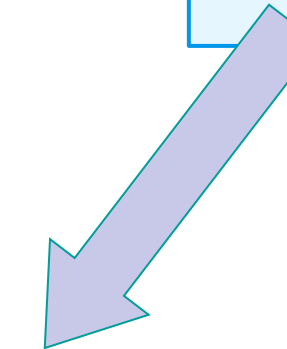
Capital letter = Master copy and lower case = copy

Type	Short ID	Long ID
Presentation	U_<X>	U_Name
Execution	E_<X>	E_Name
Installation	I_<X>	I_Name
Data	D/d_<X>	D_Name
Technical	TE_<X>	TE_Name

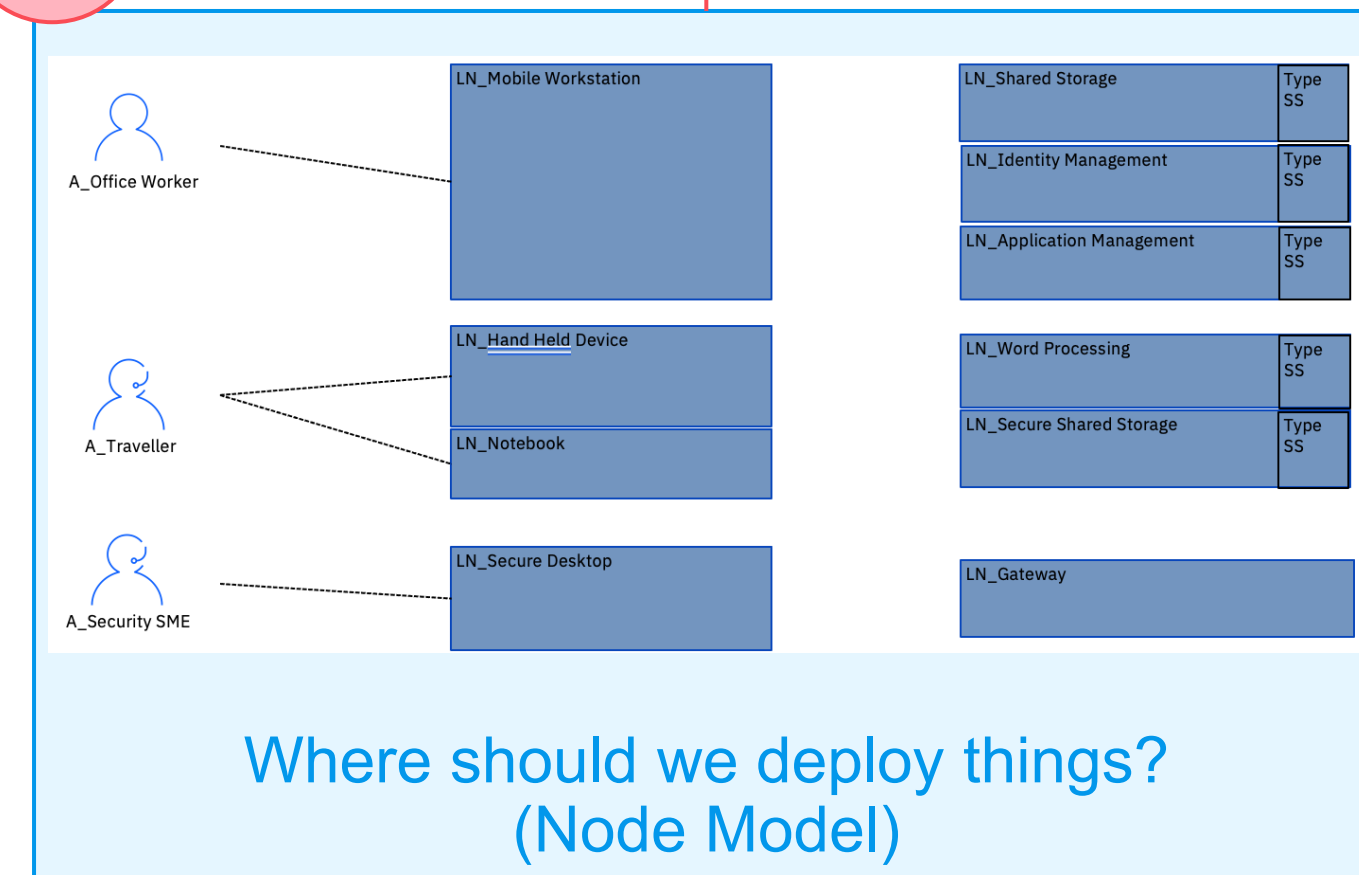




AOD, FRs, NFR's, constraints

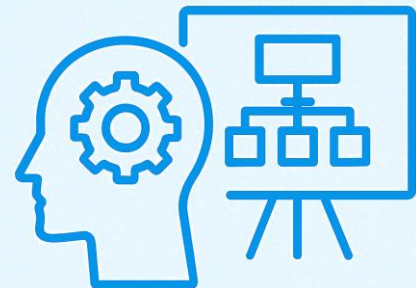


2

3

What do we need to deploy?
(Deployment Unit Model)
These contained artifacts give us a way of “building up” the
Operational Model

Where should we deploy things? (Node Model)



LL_Office (5)



LL_Remote (10M)



LL_Secure (300)



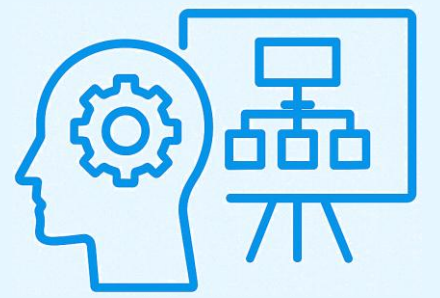
LL_Cloud (1..n)



LL_Secure DMZ (1)



Deployment Units represent all the bits required to bring an application component to life within the operational model



Presentation

Entry point used by actors to access a components functionality e.g. live web interface.

Data

The data used by components e.g. user profile.

Execution

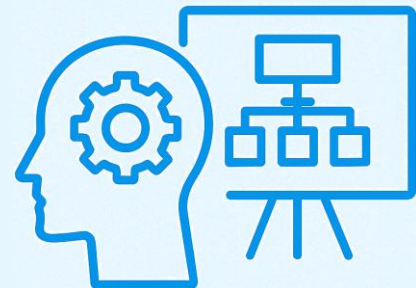
The runtime execution aspect of a component e.g. 3 instances of a word processor running in 3 different user accounts on the same mobile workstation.

Installation

Files required to install the component e.g. executable, configuration and data.



Logical Deployment Units



List deployment units for each component

Component (from the CM)	Presentation Deployment Unit	Data Deployment Unit	Execution Deployment Unit
Customer Relationship Management	U_CRM	D_CRM_Data	E_Update_CRM_Data E_Consult_CRM_Data
Stock Management	U_SMS U_Inventory	D_Inventory_Upd	E_Submit_Inv_Update E_Consolidate_Inv_Update

DUs need to be described regarding their non-functional requirements which determine their placement and grouping.

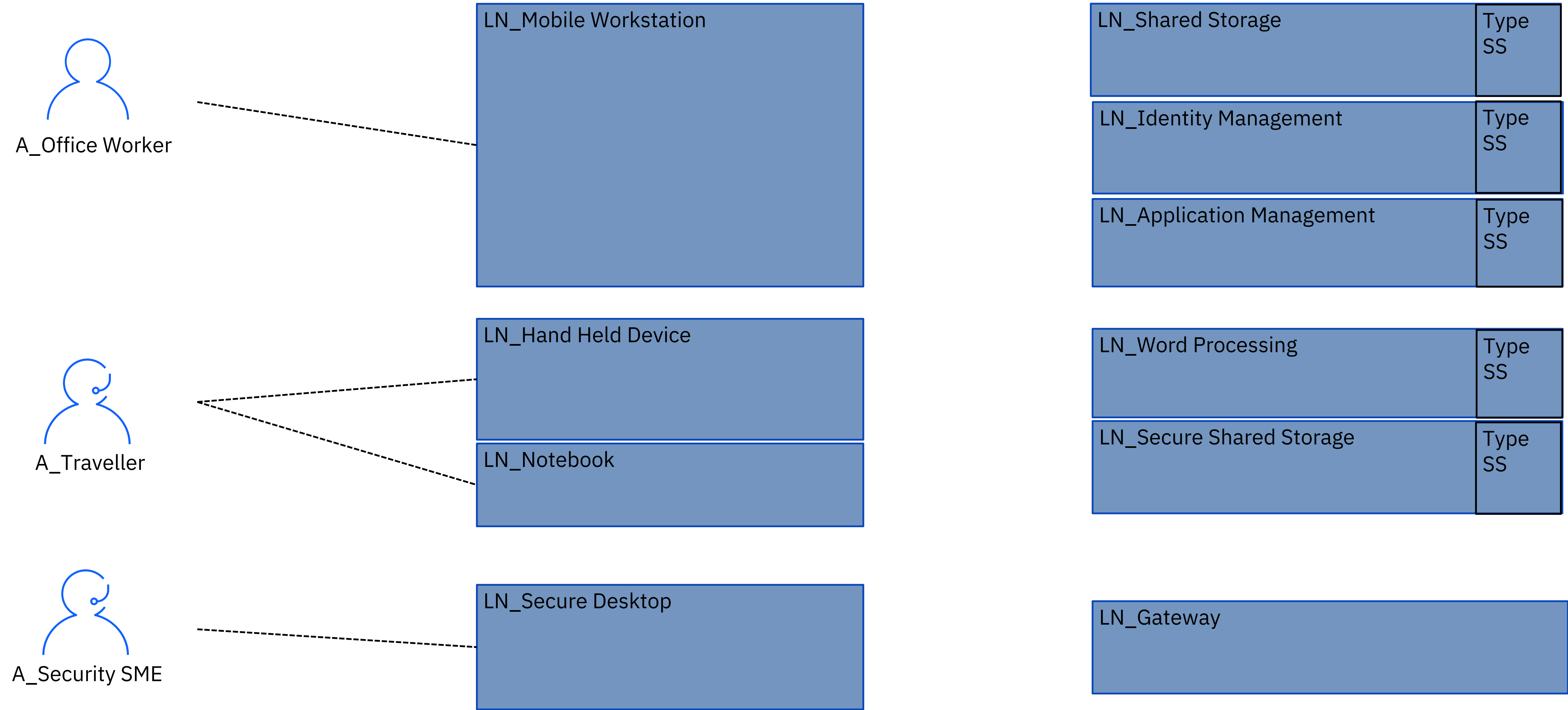
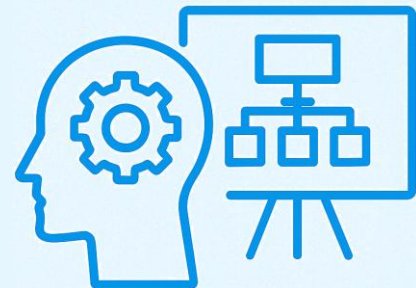
PresentationDU	Actor	IT Skill level	Location	Frequency	Type
U_Inventory	A_Inventory_Check	Can be trained	Mobile	N/A	N/A
U_SMS	A_SMS	N/A	N/A	Weekly	Batch

Data DU	Scope	Volatility	Size	Lifetime	Master or Copy
D_inventory	Store	High	Small	<= 1 day	N/A
D_Cat	Entire product range	Low	Large	Long lived	M

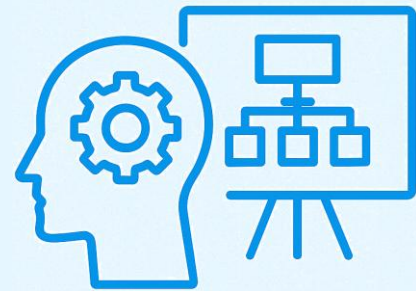
Execution DU	Frequency	Processing Load
E_Submit_Inv_Update	Daily (once for every store)	Light
E_Consolidate	Daily	Heavy



Logical Node Model



We decide where the Deployment Units WILL go Following One of Two Paths



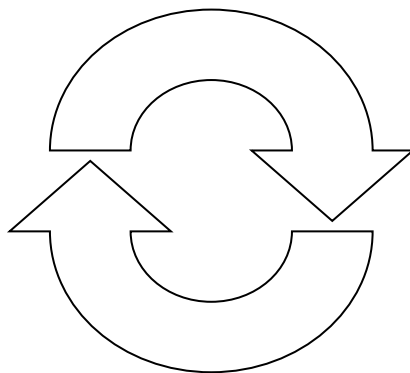
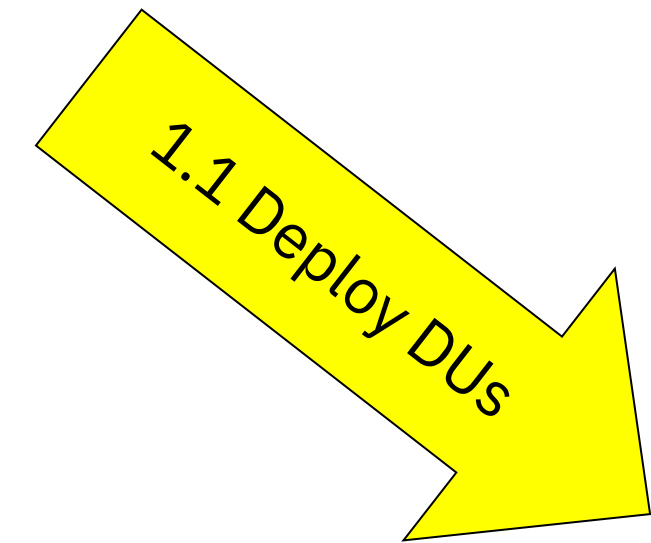
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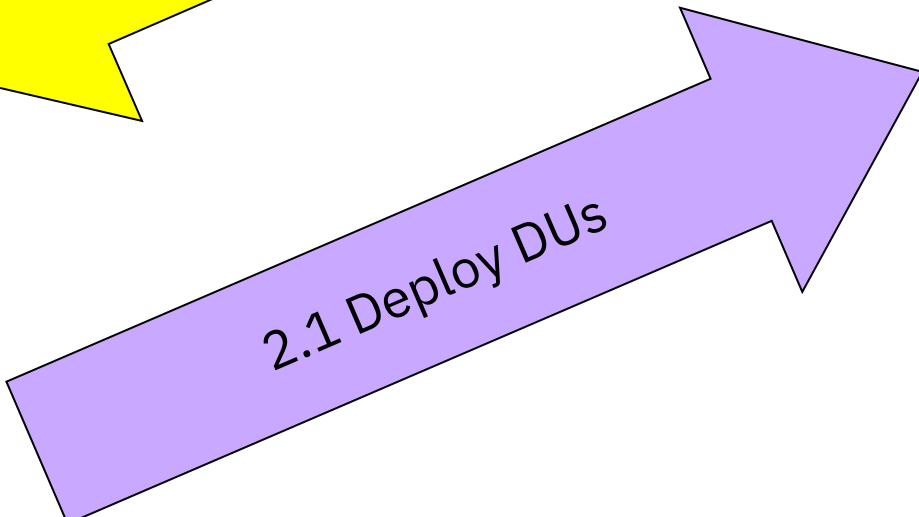
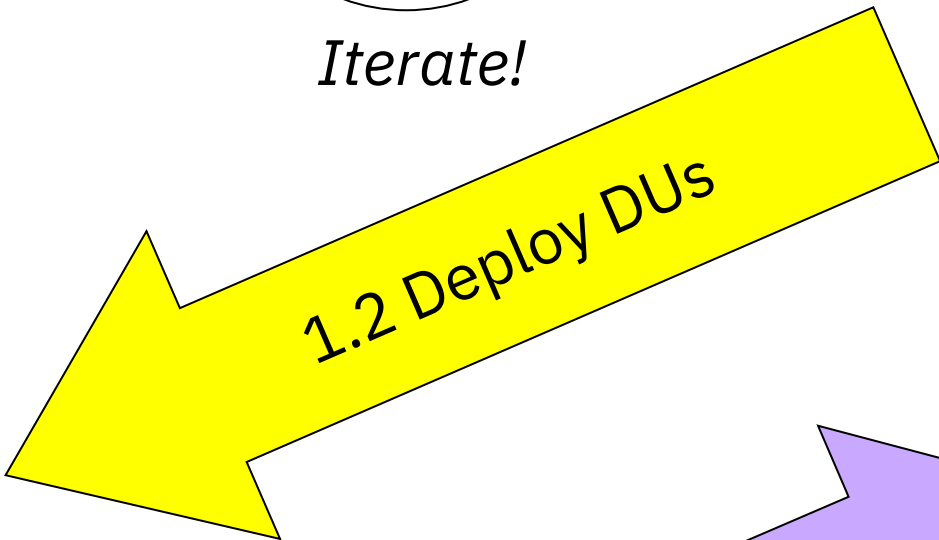
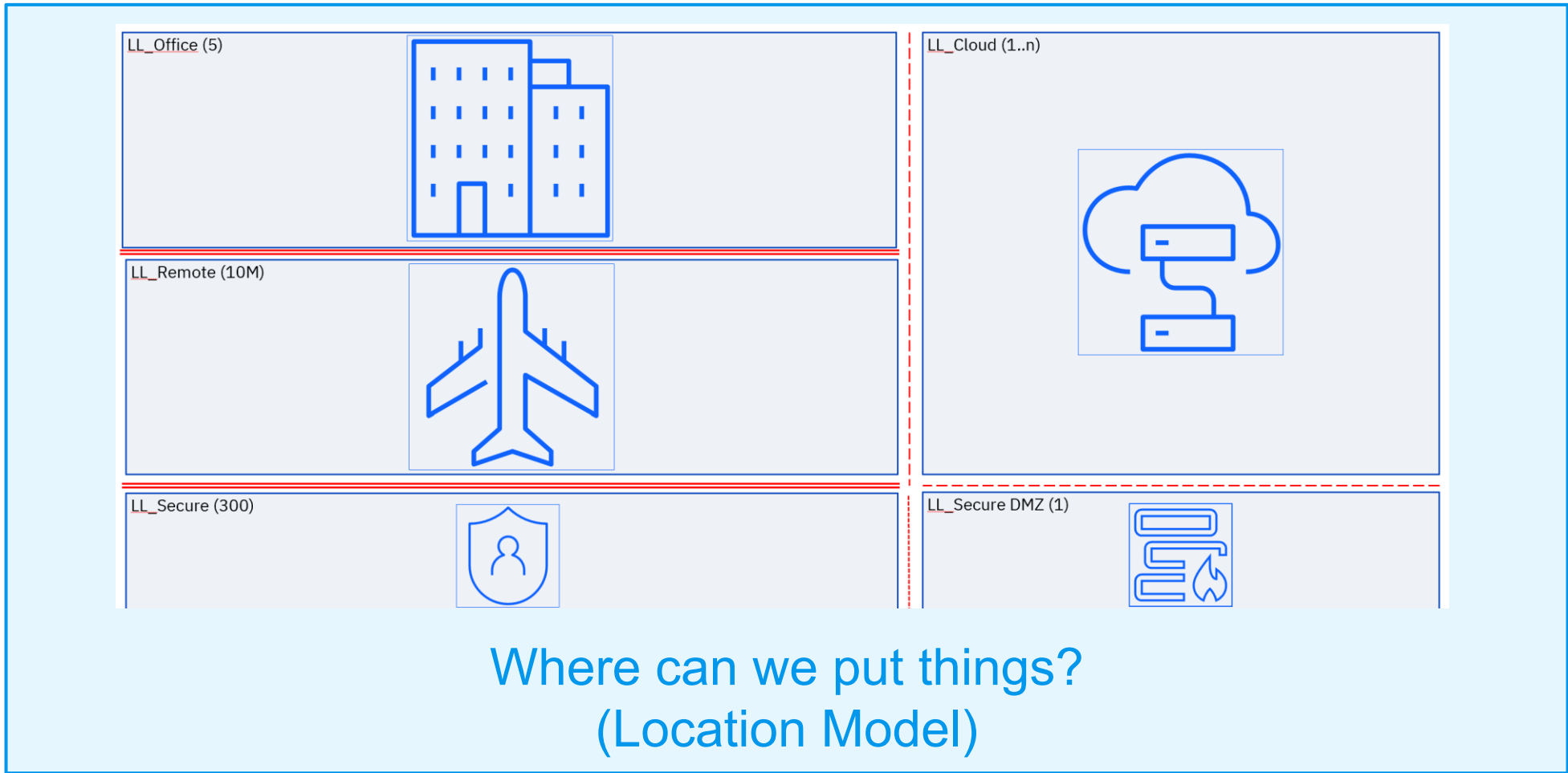
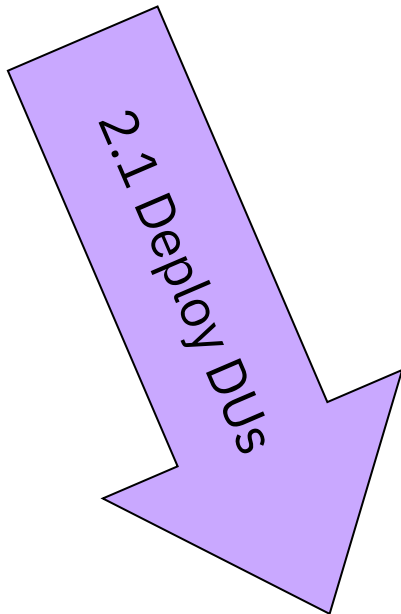
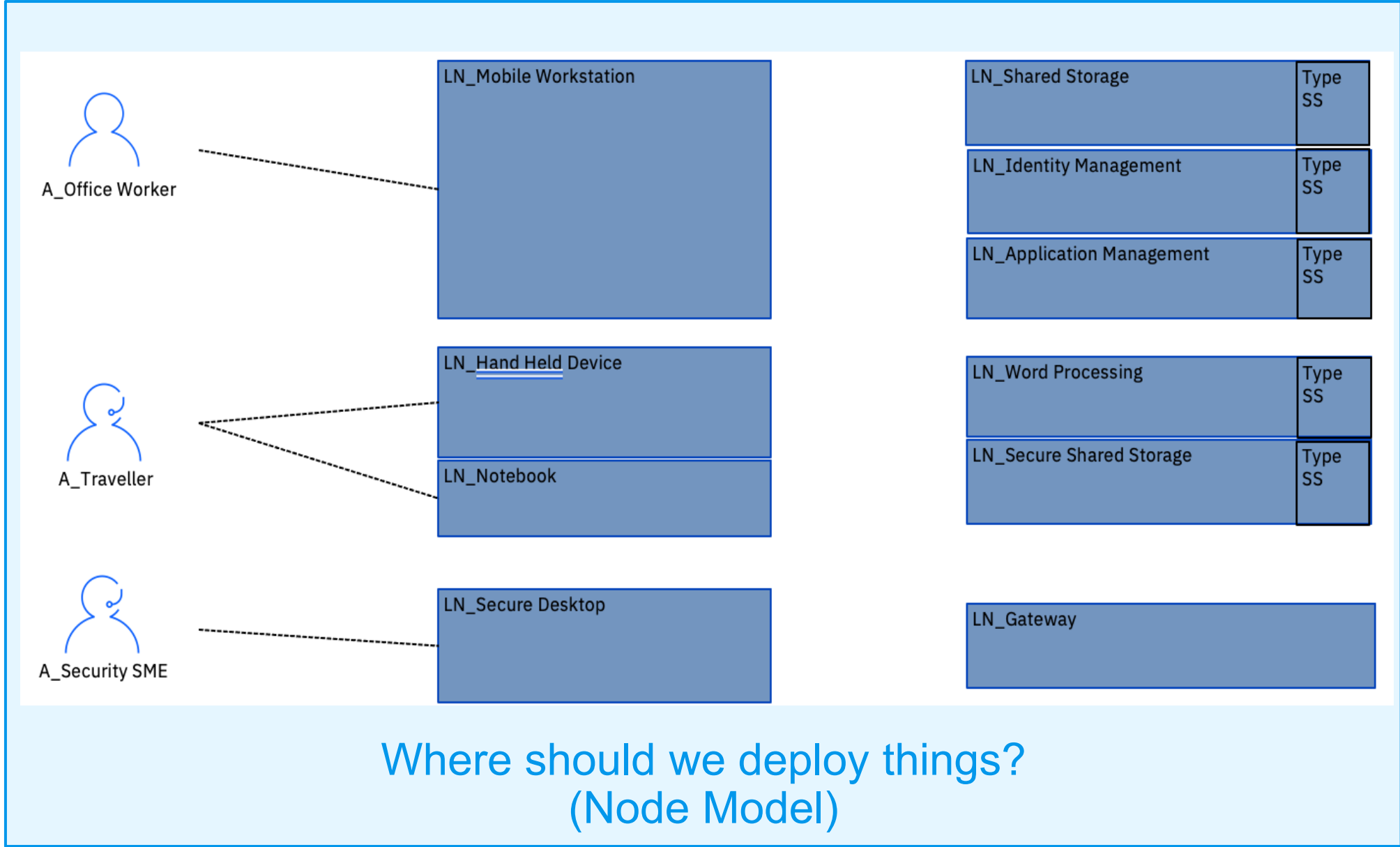
Execution DU	Frequency	Processing Load
E_Submit_Inv_Update	Daily (once for every store)	Light
E_Consolidate	Daily	Heavy

What do we need to deploy?
(Deployment Unit Model)

Path 1 using an asset where DU's are associated with a predefined node model and nodes are deployed into their locations. We will use this approach for the worked example.

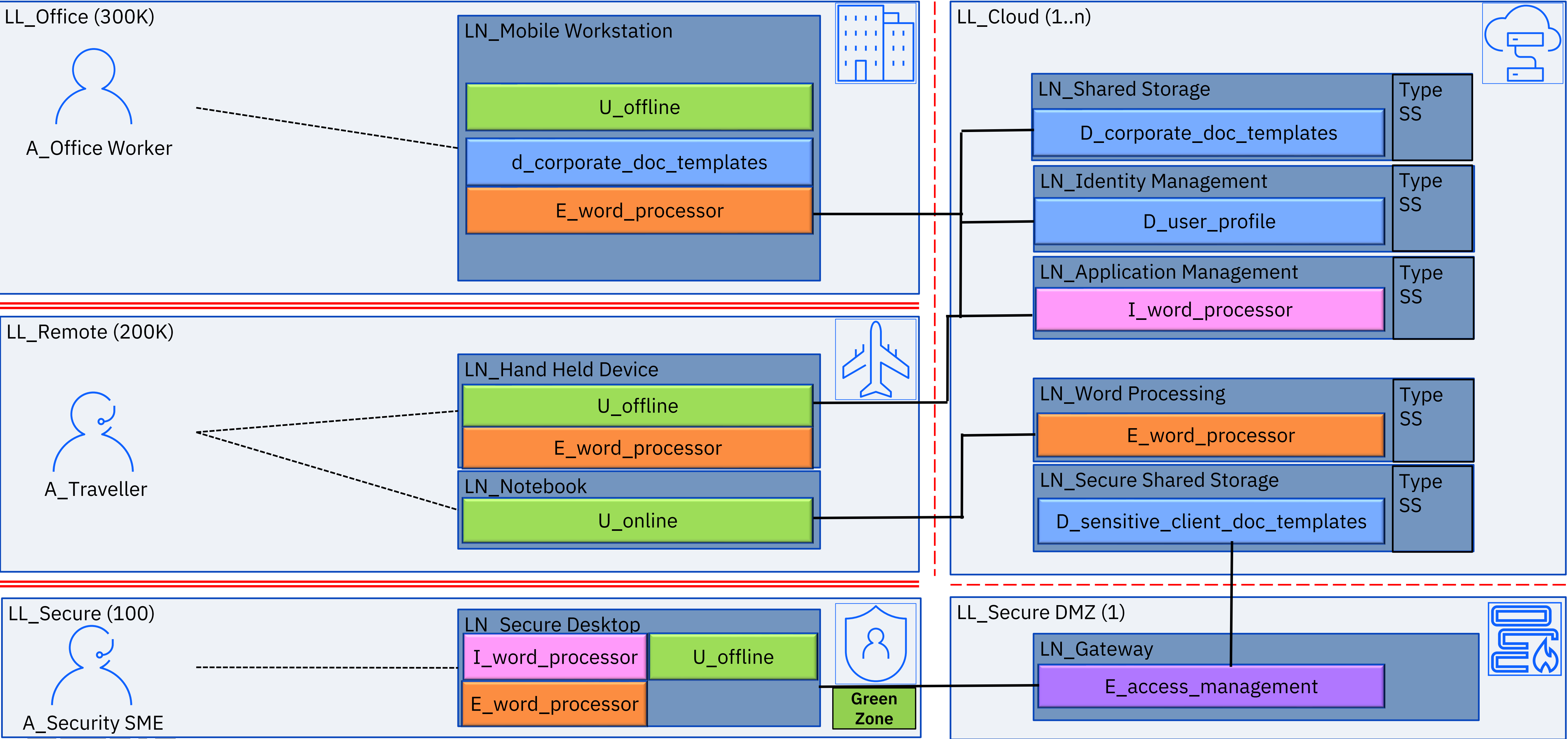
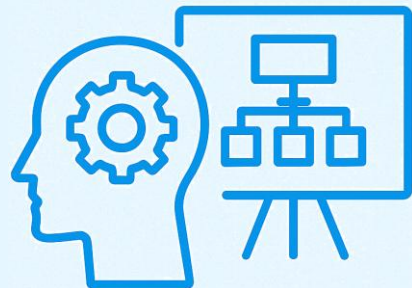


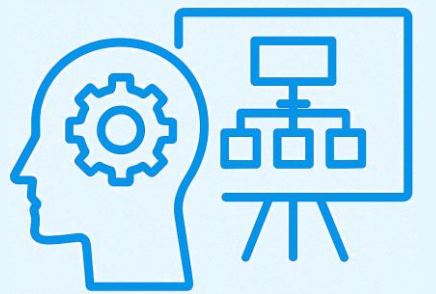
Iterate!



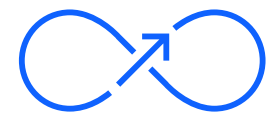
Path 2 nodes are formed at locations that contain DU's

Logical Operational Model - shows the static relationships between nodes, DUs, connections and interactions on a geographic background





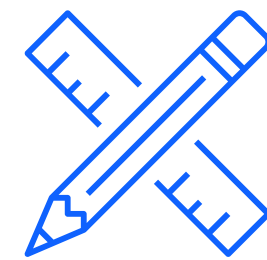
The Operational Modelling technique consists of multiple steps



1. Understand the non-functional requirements

What NFRs need to be met?

1M transactions per hour



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Customer Relationship

Application



3. Conceptualize the Logical Operational Model (LOM):

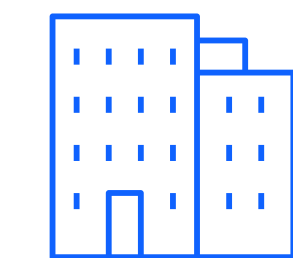
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- Determine what logical technical components are required and where they will be deployed to support the logical application components

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Load balancer

Database

Logical



4. Transform into the Prescribed Operational Model (POM):

Determine real technologies to deliver all of these specifications.

SAP

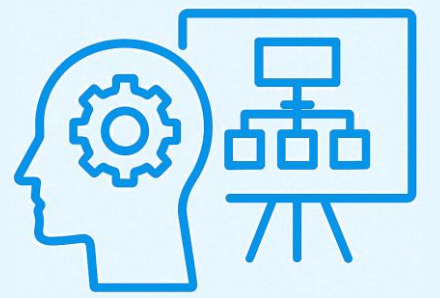
Message Broker

F5 BIG-IP LTM-4200v

DB2

Prescribed

The LOM documents the placement and interaction of the application's DUs onto nodes distributed across the system's locations



Identifying potential locations and nodes where deployment units might end up being placed, and establishing where the system's actors will be located

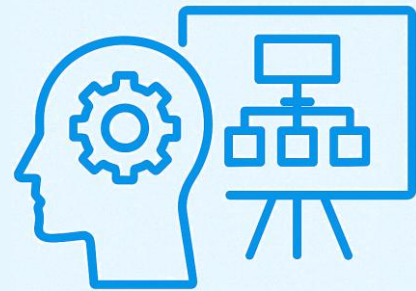
Placing deployment units onto application logical nodes in locations, based on a variety of considerations such as service management

- Presentation DUs
- Data DUs
- Execution DUs
- Installation DUs

We derive connections from the relationships (interactions) between deployment units (Confirming the design using architecturally significant walk-throughs)



Step 1: The overall structure of the LOM is largely based on the locations over which the solution will be deployed.

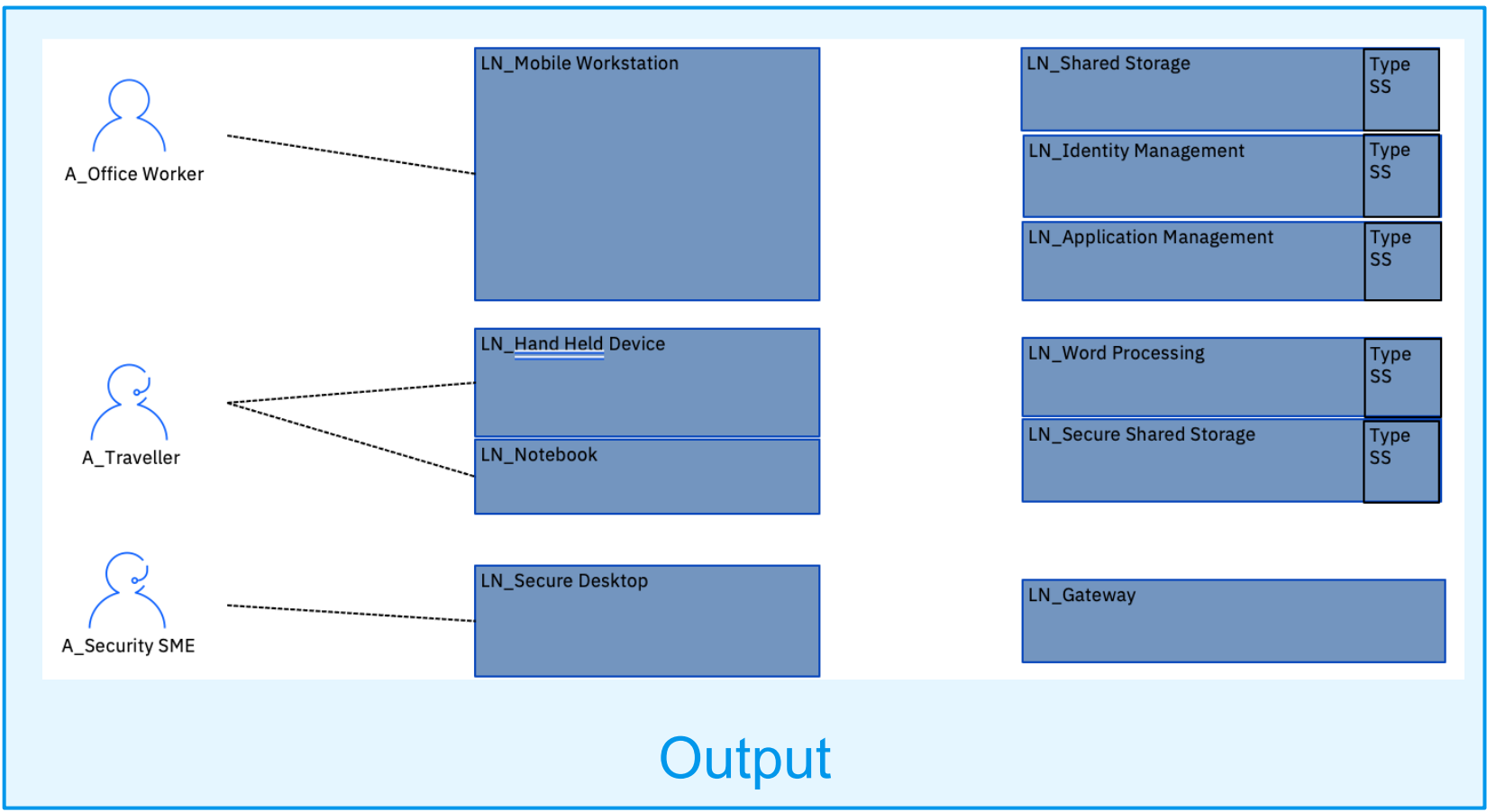
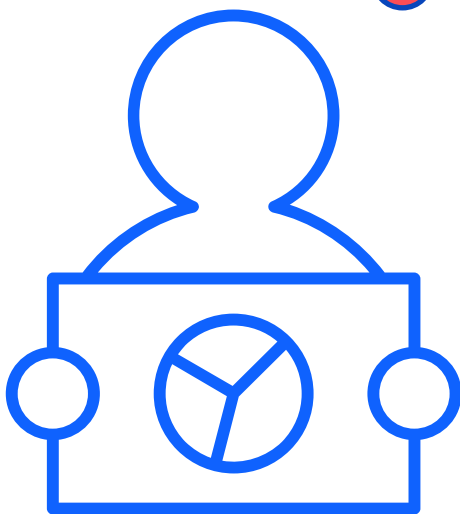
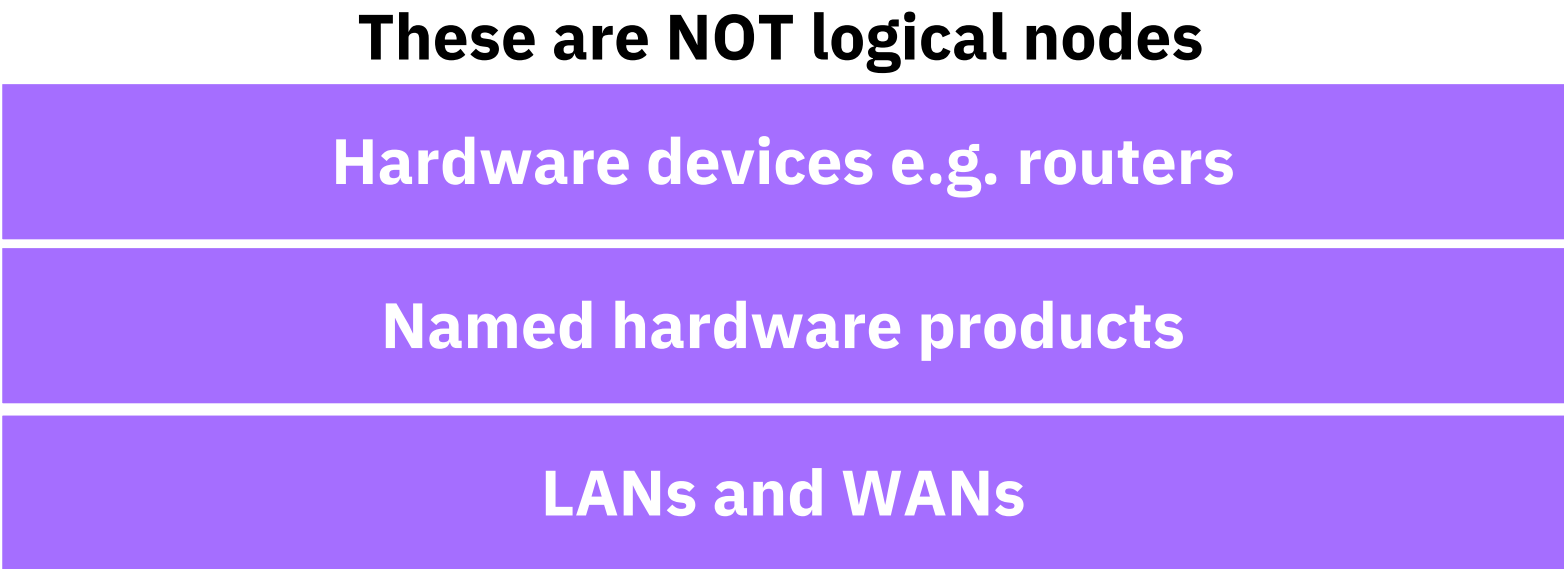
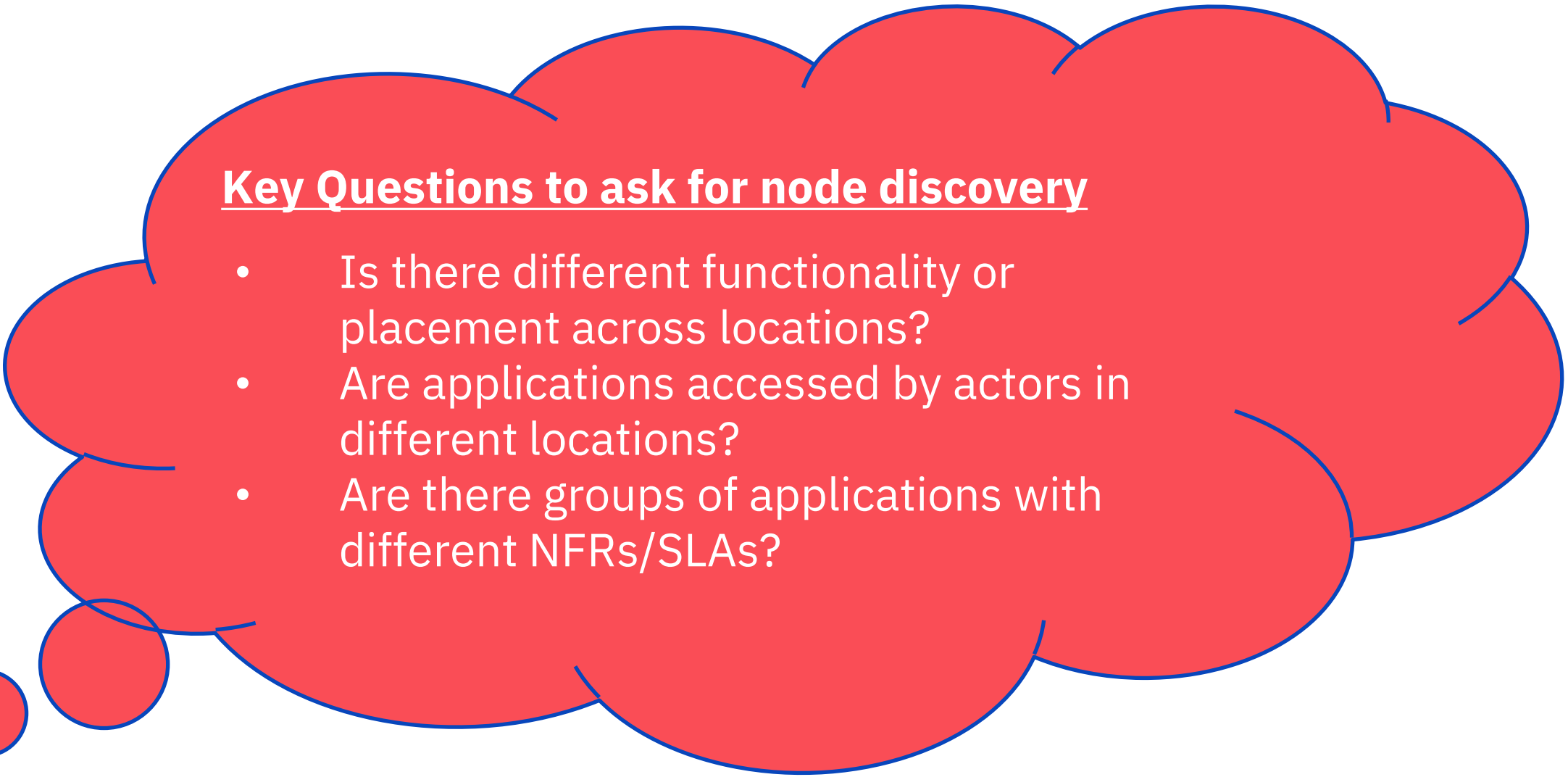
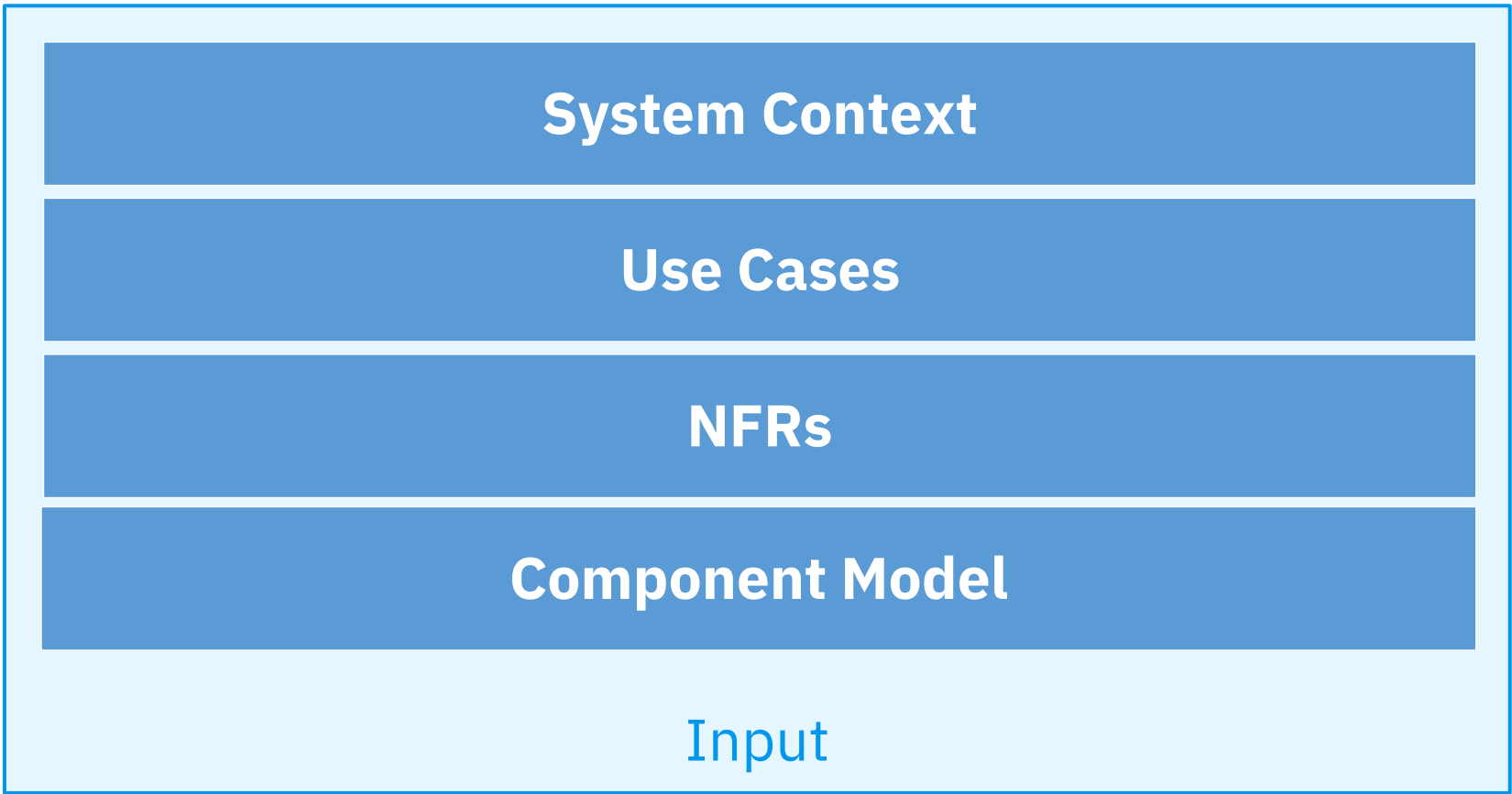
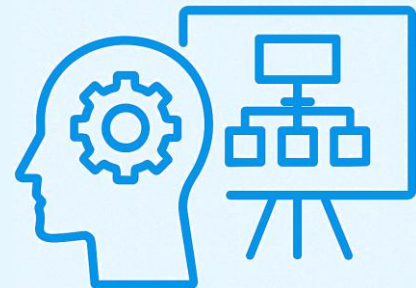


The location is based on the information found in:

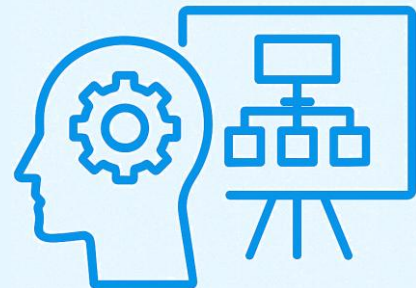
- Business Roles & Locations
- System Context Diagram
- Current IT Environment



Step 2: Then identify Candidate Logical Nodes



Step 3: We have to understand what it is we need to deploy, via the DEPLOYMENT UNIT Model



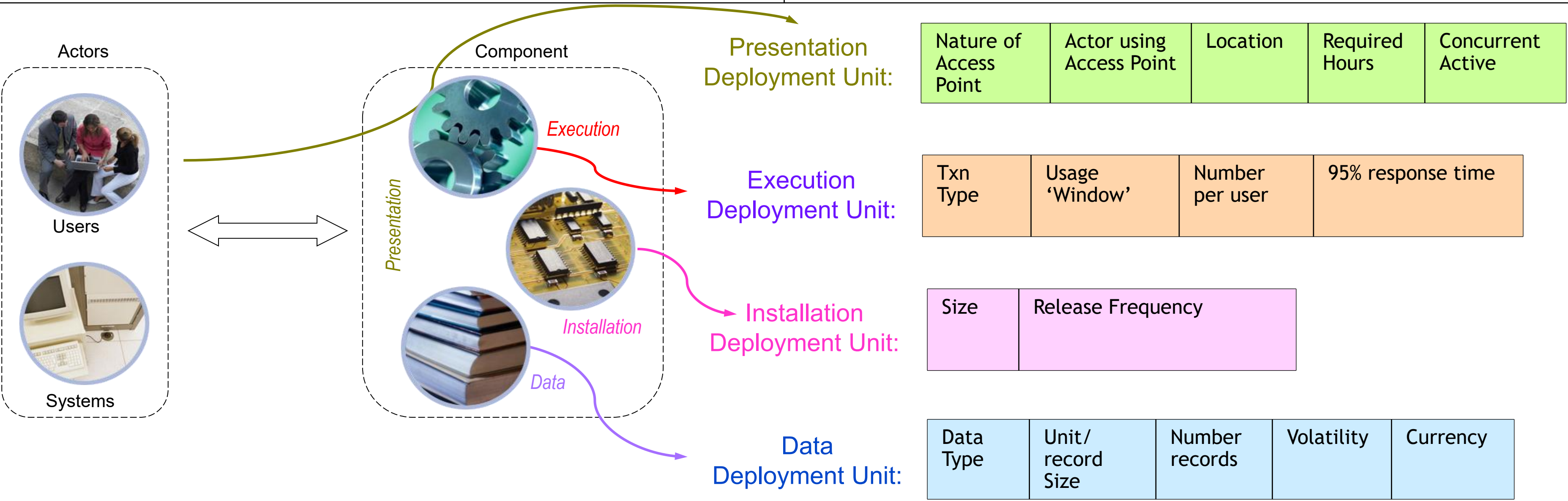
DUs give us a mechanism for:

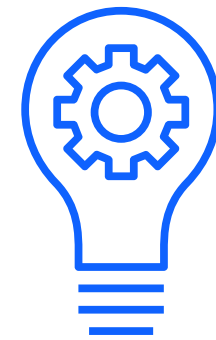
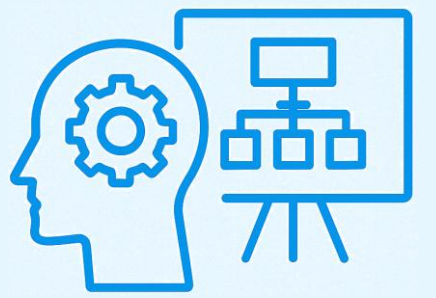
- Understanding the nonfunctional requirements placed on the system’s components
- Deciding where best to place the various aspects of the system’s components
- Tracing the system’s requirements to its design

The nonfunctional requirements of the component’s interactions are described in terms of DU – DU relationships.

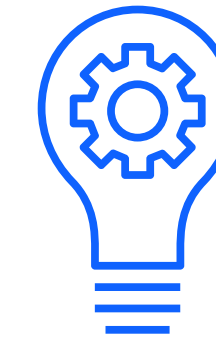
Typical relationships analyzed include:

- Presentation ↔ Execution
- Execution ↔ Execution
- Execution ↔ Data
- Presentation ↔ Data





Time for a Question

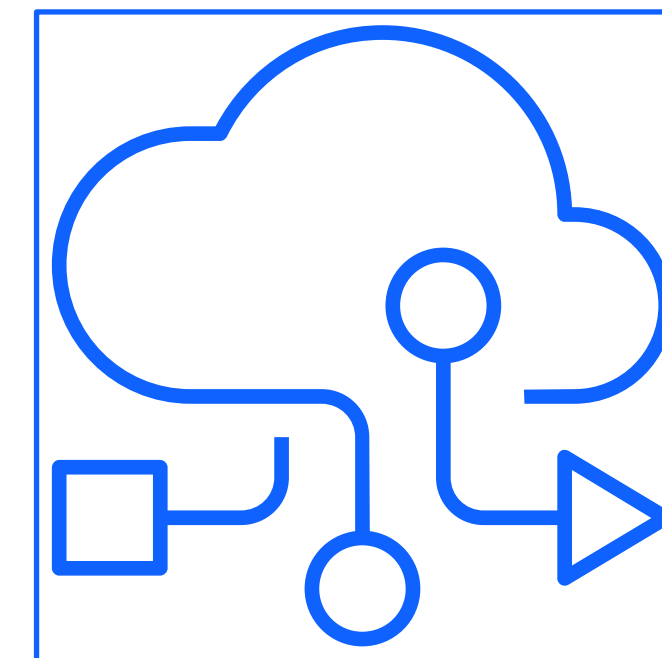
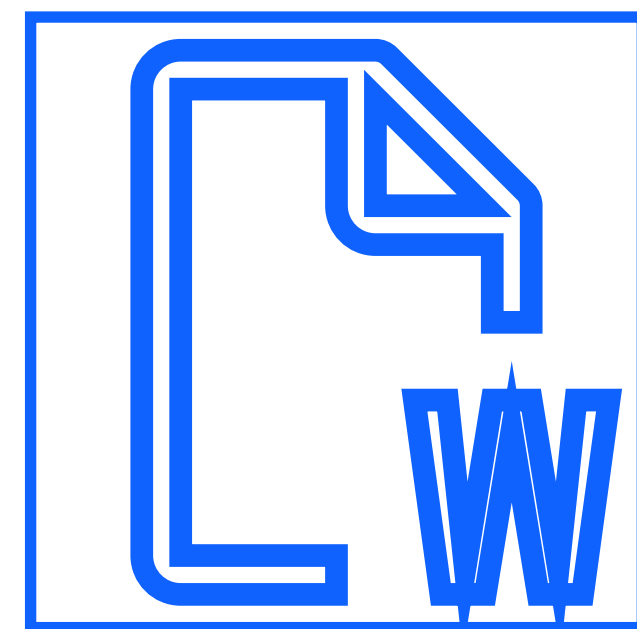
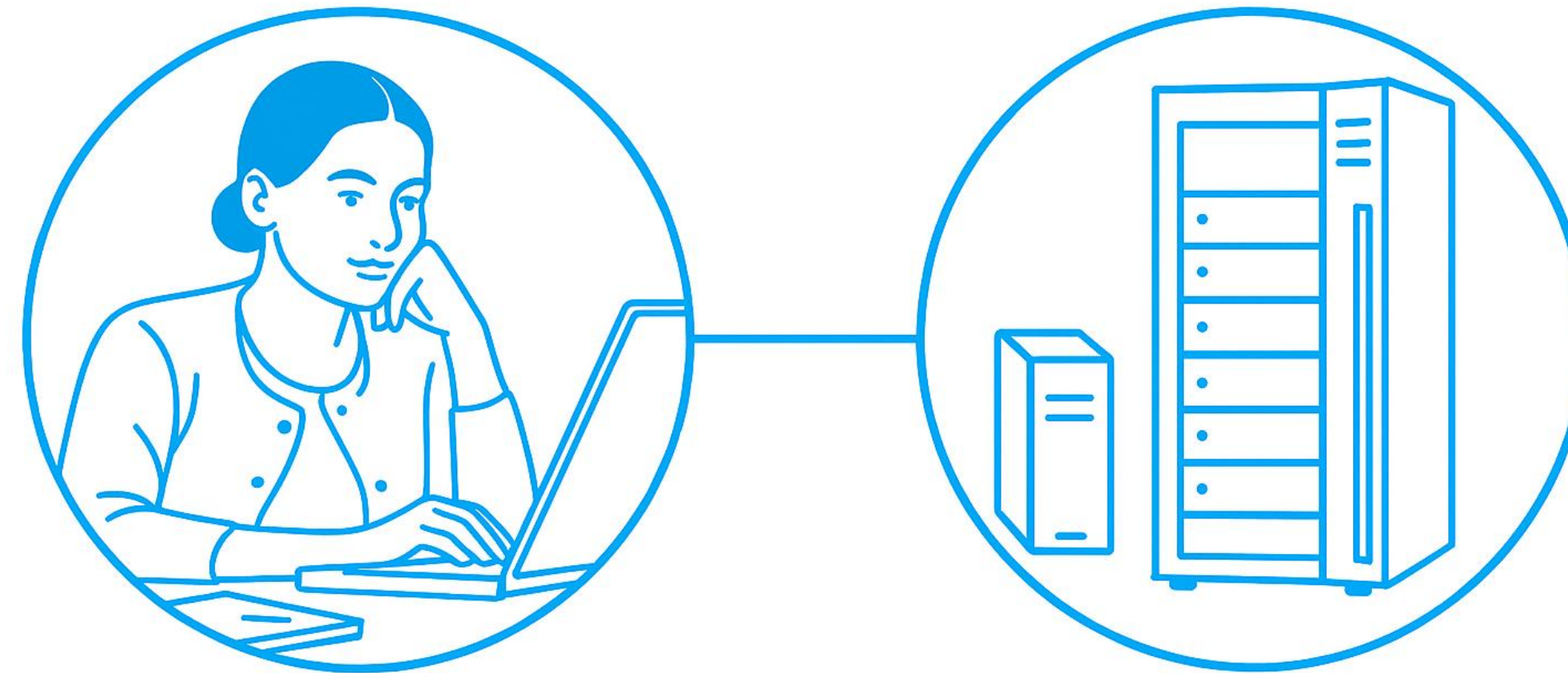
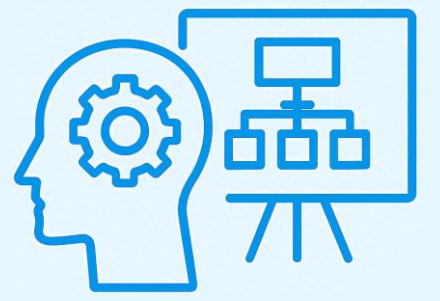


The typical views of an Operational Model include the Prescribed Operational Model, and the:

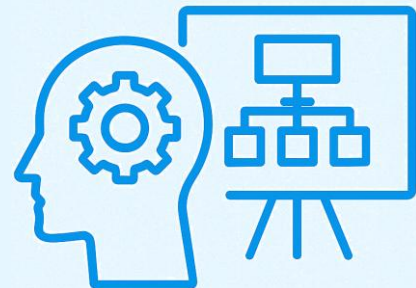
1. Logical Operational Model
2. Component Operational Model
3. Technical Operation Model
4. Deployment Operation Model

Example:

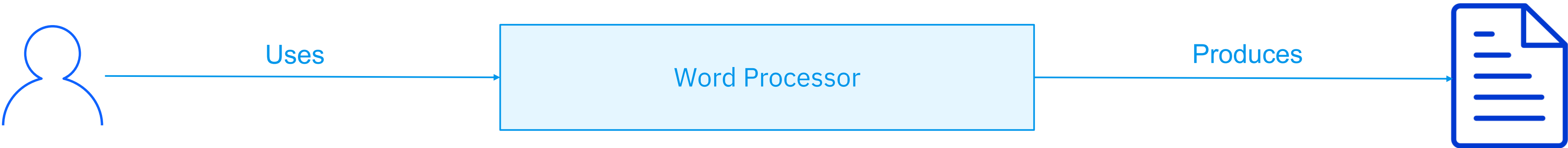
Where and how do we “deploy” a Word Processor ?



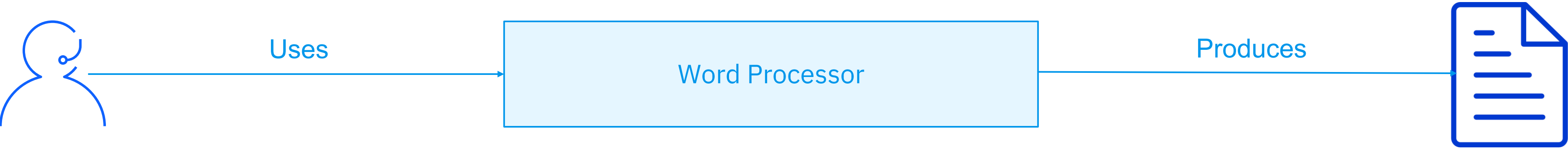
What are the use cases that need to be met ?



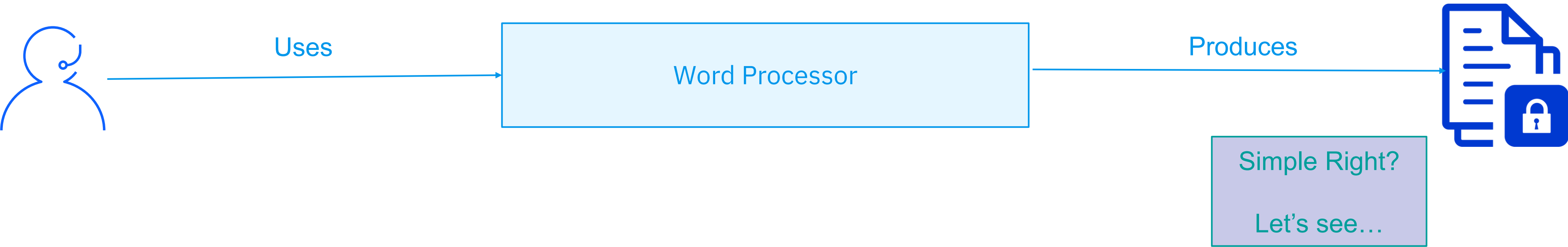
Story 1: Sales Manager working from an IBM office creates a statement of work using an IBM approved template.



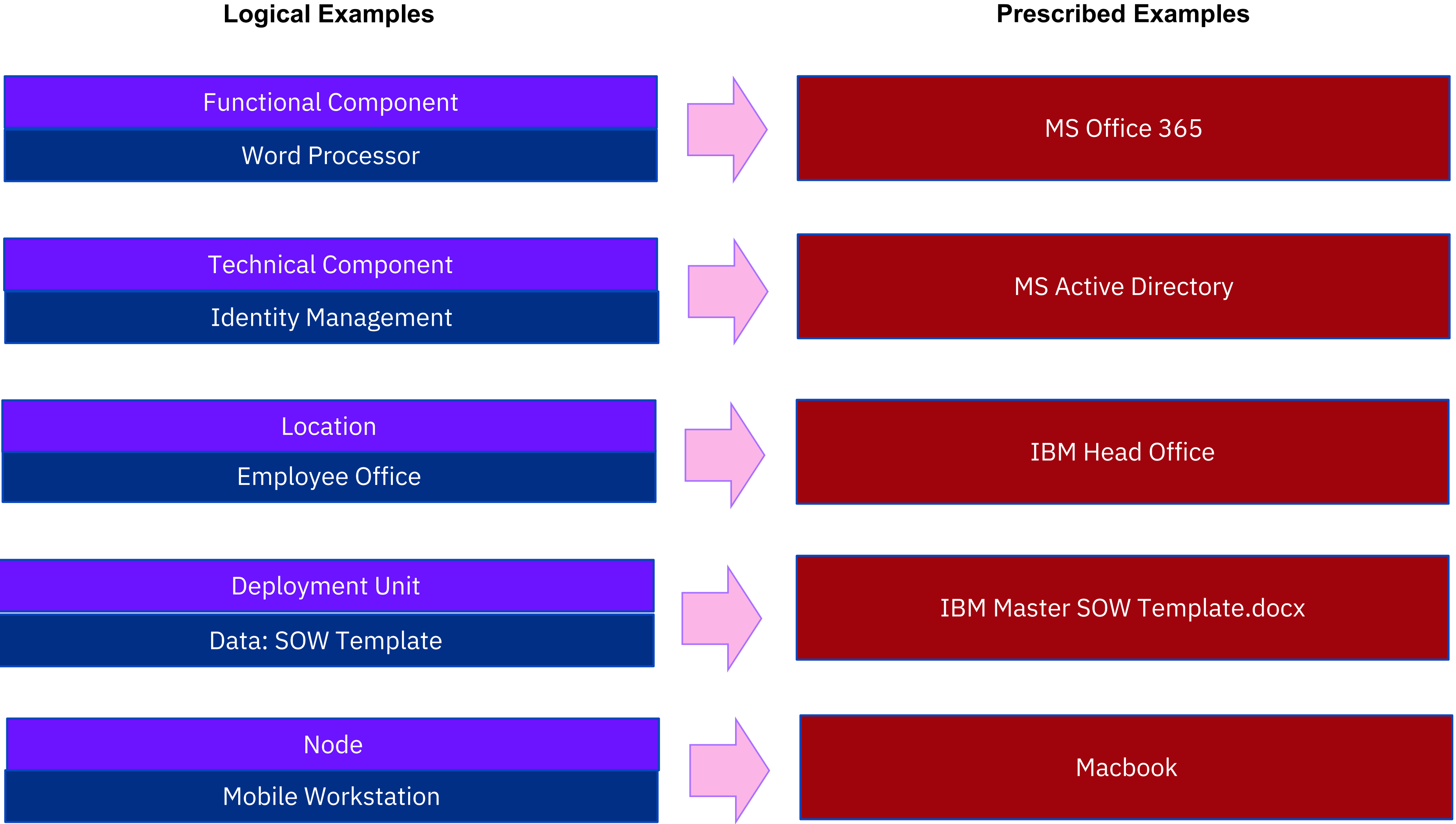
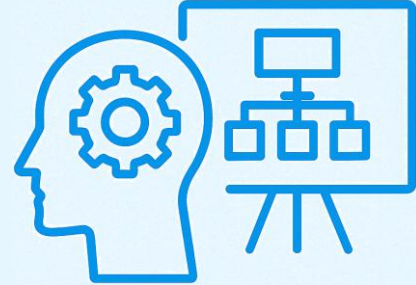
Story 2: Project Manager while traveling creates a Project Change Request using an IBM approved template.



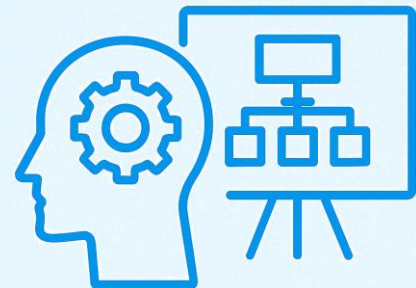
Story 3: Architect working from an IBM office in the public sector creates a confidential design using a template from an IBM asset that is not allowed to be stored outside of local country.



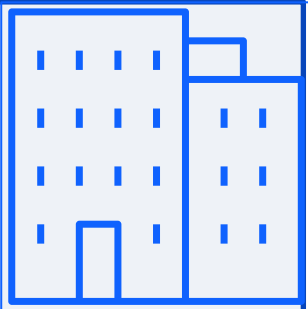
First we model the landscape from a logical perspective and then from a prescribed perspective



Word Processor Example - Logical Location Model



LL_Office (300k)



LL_Remote (200k)



LL_Secure (100)

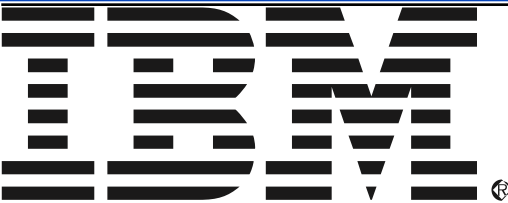


LL_Cloud (1..n)

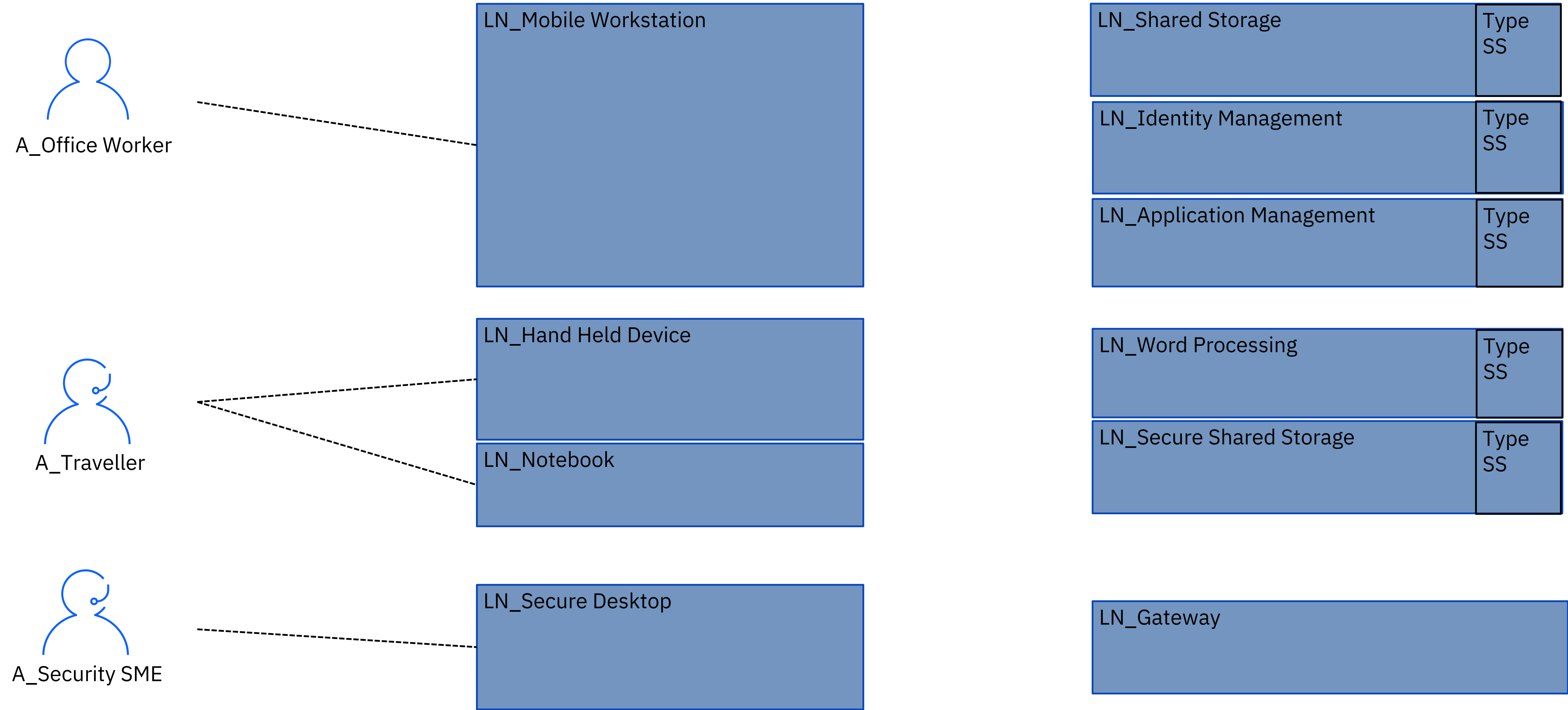
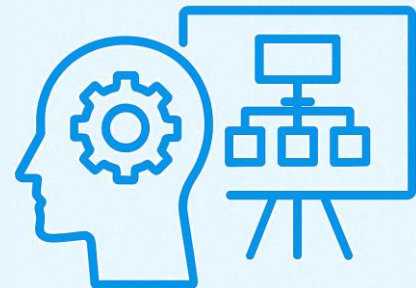


LL_Secure DMZ (1)

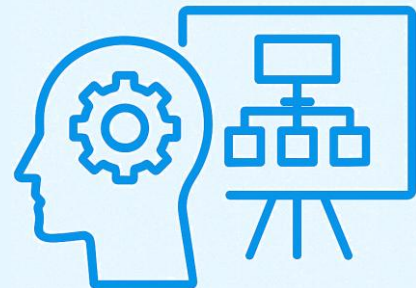




Word Processor Example - Logical Node Model



There are many ways of deploying a single component into a simple system



Everything on the PC...

Option 1

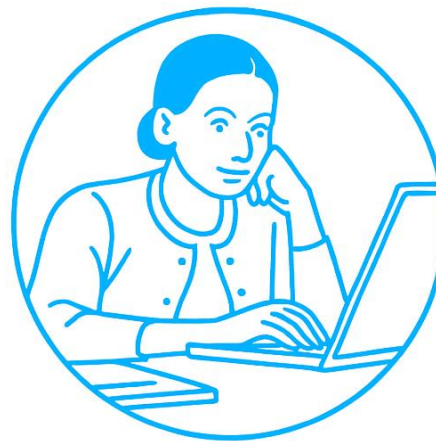


Presentation
Execution
Data
Installation



Word, Running on
VM, standalone

Option 2



Presentation
Execution



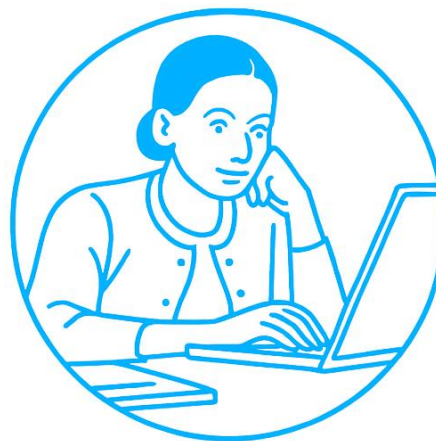
Data
Installation

Word, running on Win10, with remote file serving

In this case we have 16 options in minimum
to place DUs. In general this is:
(#components * #locations)^#DUs

How do we do this for complex systems in a
structured manner?
Operational Modeling

Option 3



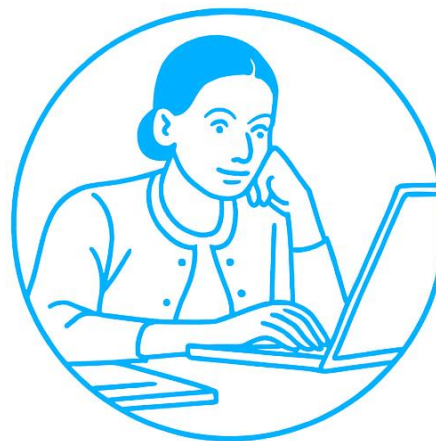
Presentation
Data



Execution
Installation

Word, running on Citrix, with local data

Option 4

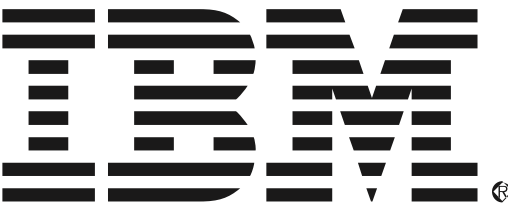


Presentation



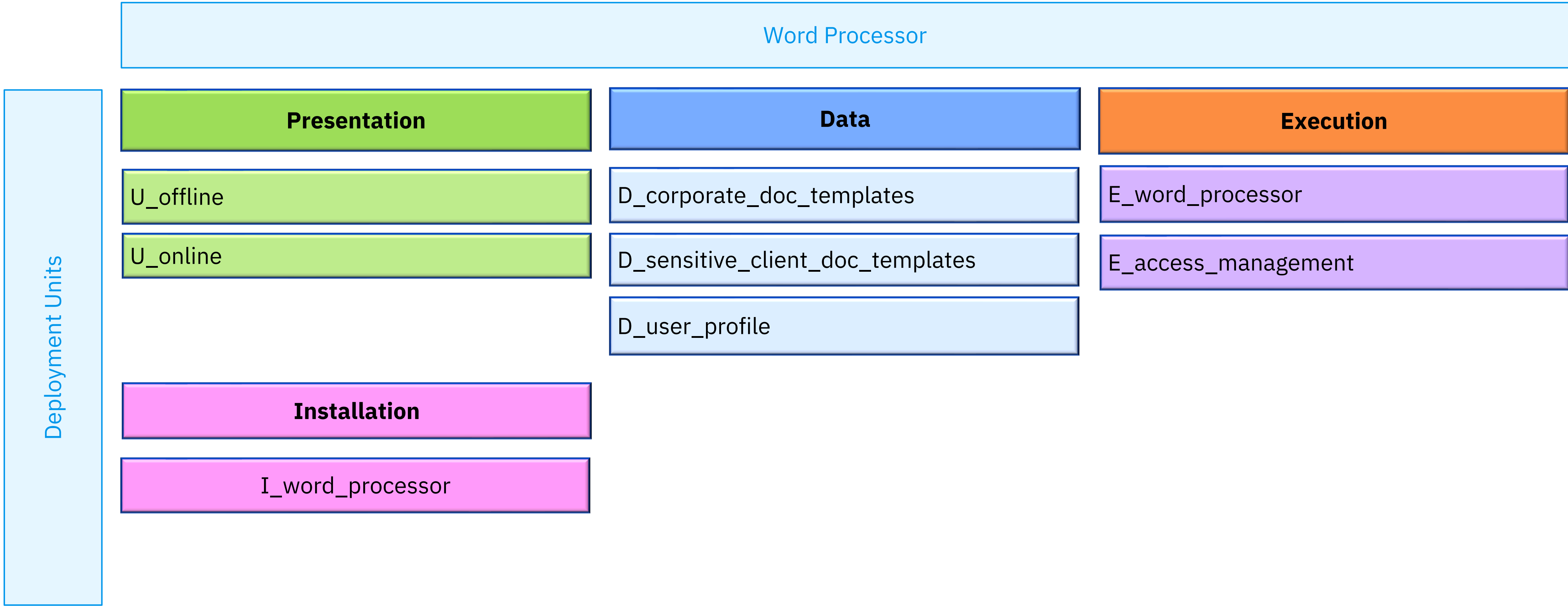
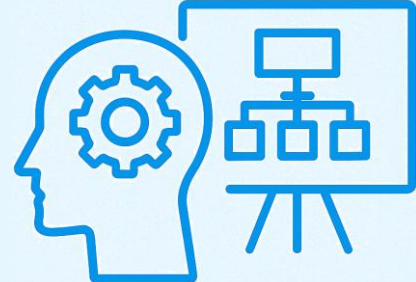
Execution
Data
Installation

Word, running in thin client with remote file serving



Everything on the Server...

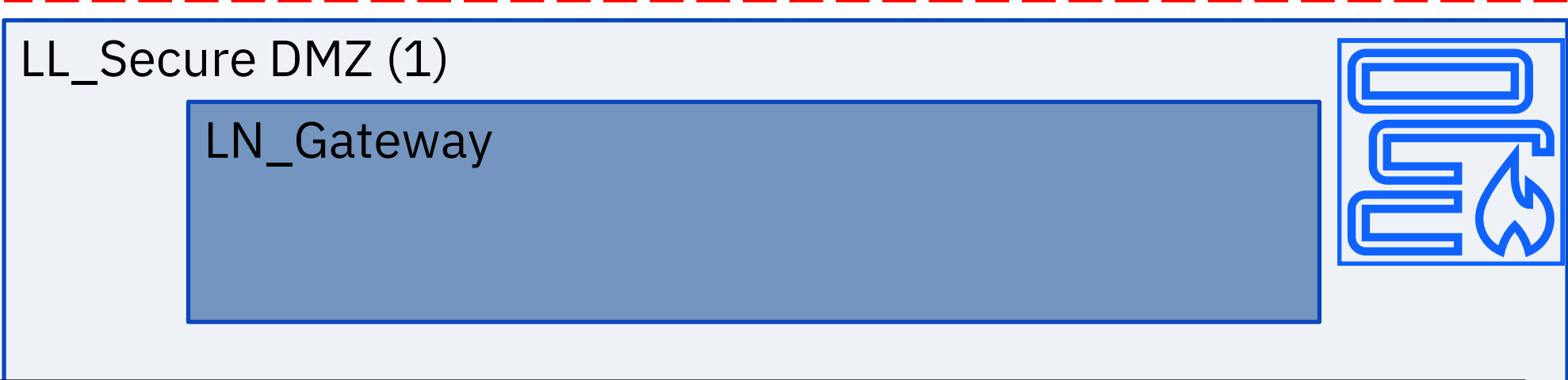
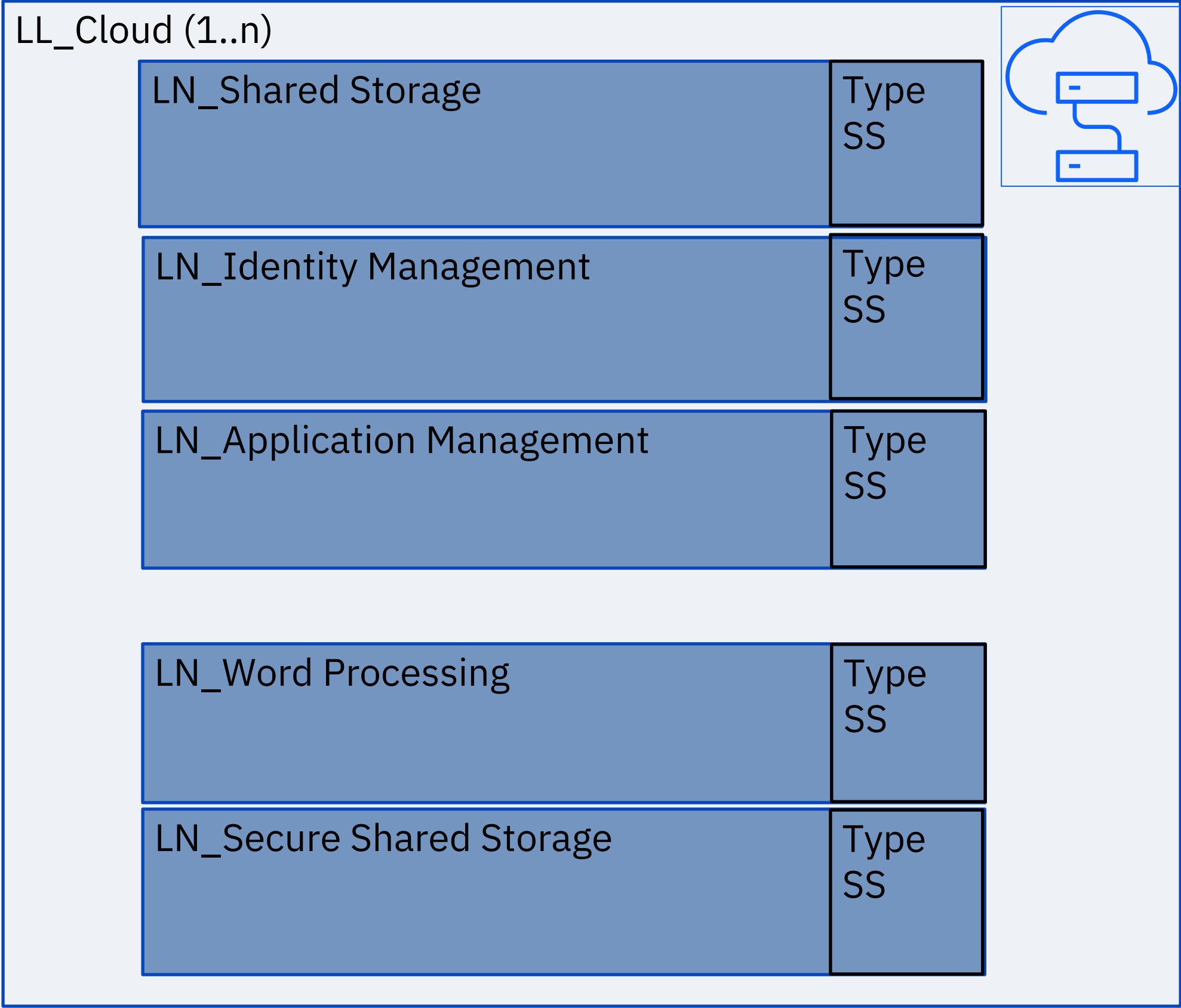
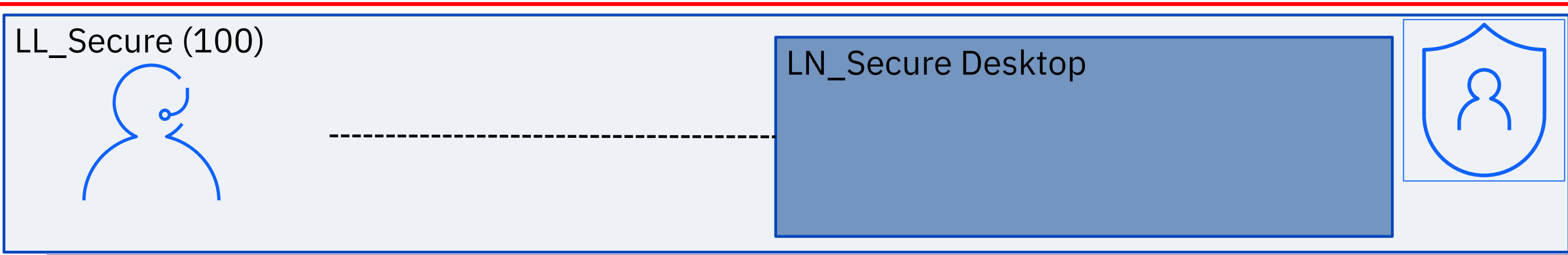
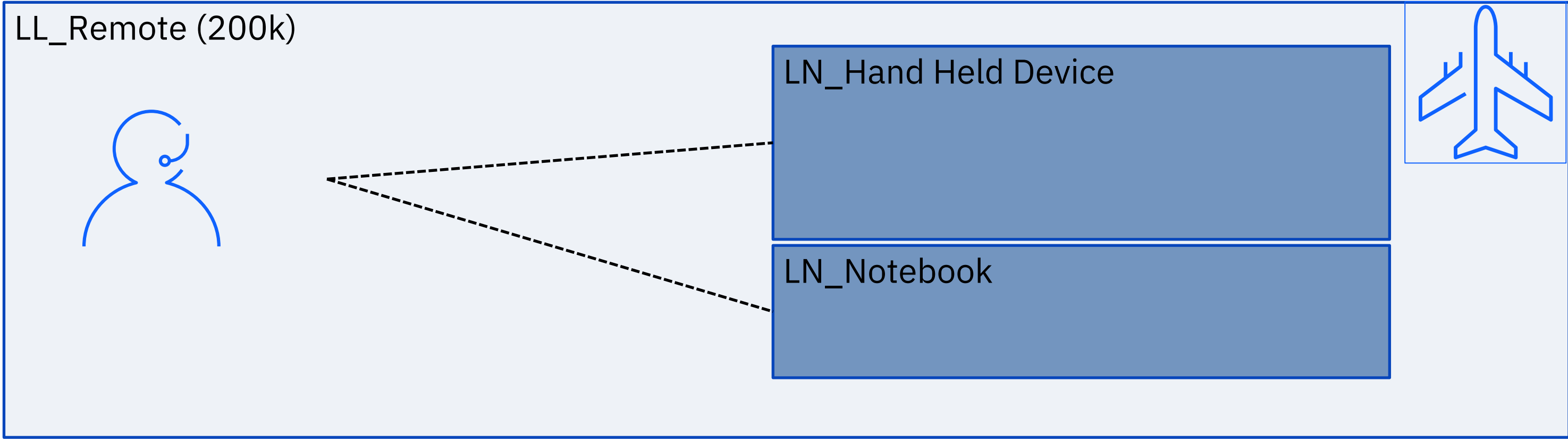
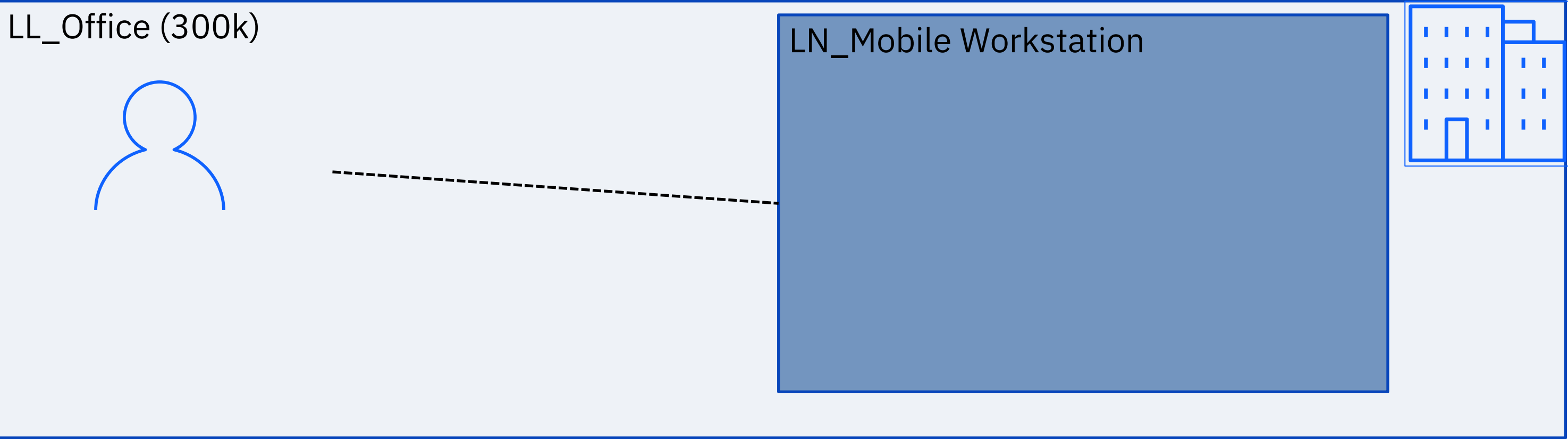
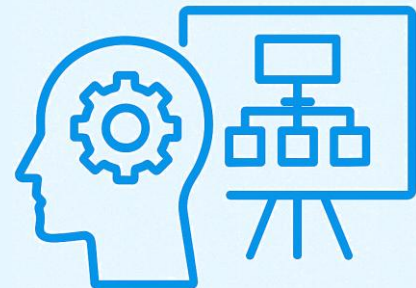
Based on our use cases we get these deployment units?



Things to consider

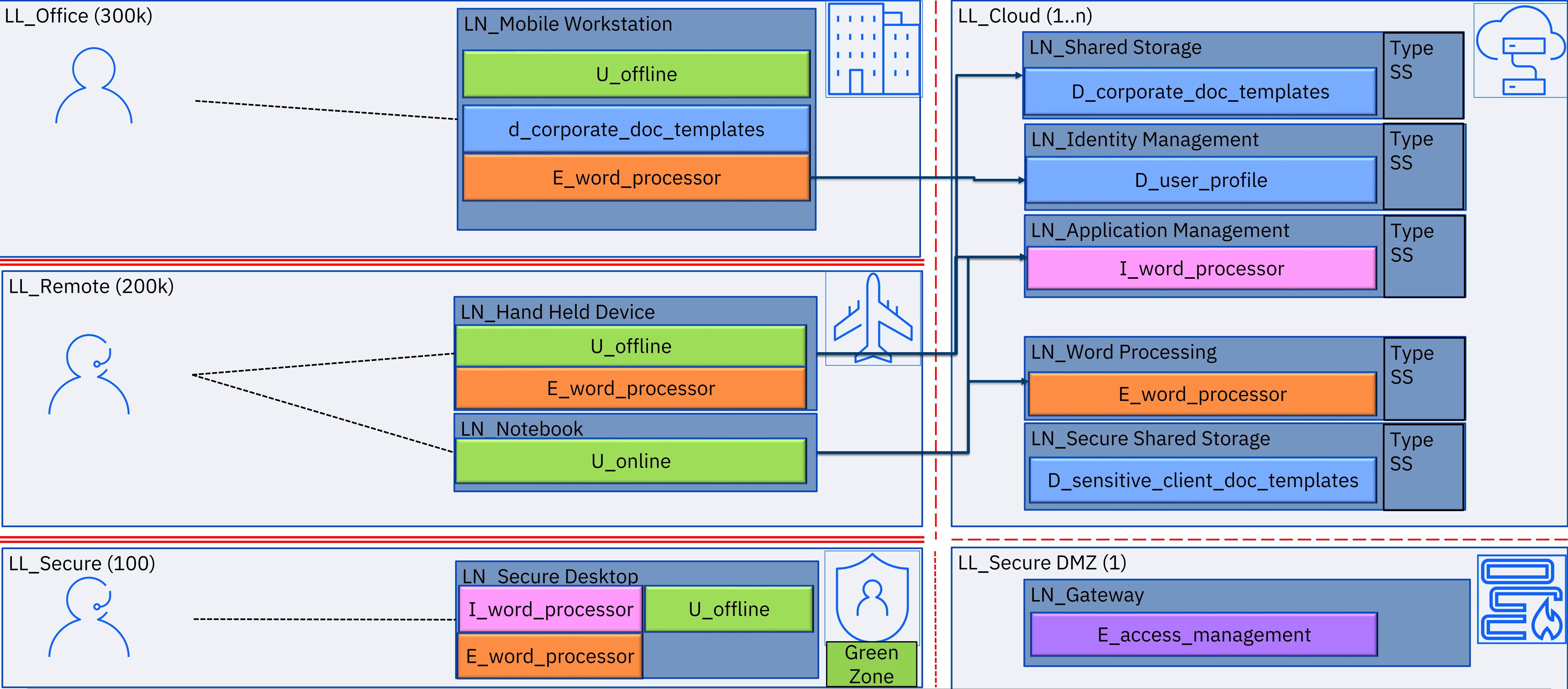
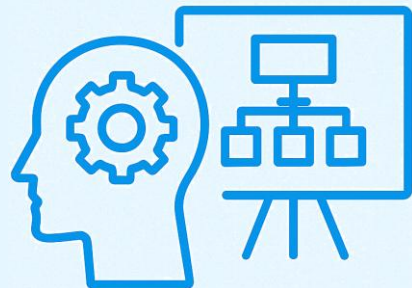
Worked Example – Word Processor Logical Operational Model Reference

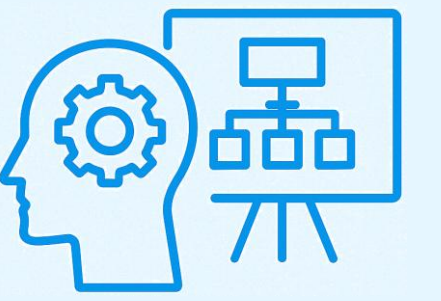
Architecture Asset Example showing existing nodes



Worked Example – Word Processor Logical Operational Model

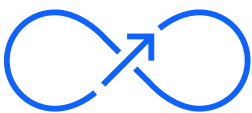
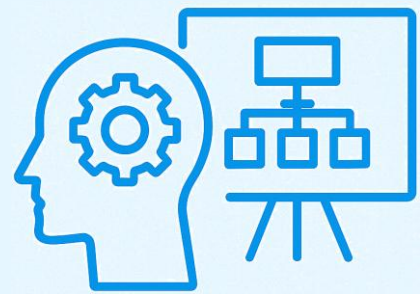
Place Deployment Units onto Nodes Following Asset





Break

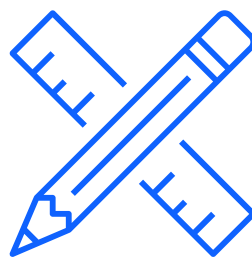




1. Understand the non-functional requirements

What NFRs need to be met?

1M transactions per hour



2. Conceptualize from an application perspective (ALOM):

Identify application level nodes and their placement.

Note: Application Logical Operational Model (ALOM) is not discussed in this lecture – see the Operational Modelling class for details on ALOM.

Finance

Warehouse Management

Customer Relationship

Application



3. Conceptualize the Logical Operational Model (LOM):

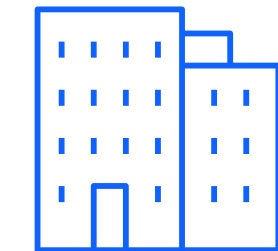
- Determine where all of the logical application components are deployed.
- Determine what logical technical components are required and where they will be deployed to support the logical application components

ESB

Load balancer

Database

Logical



4. Transform into the Prescribed Operational Model (POM):

Determine real technologies to deliver all of these specifications.

SAP

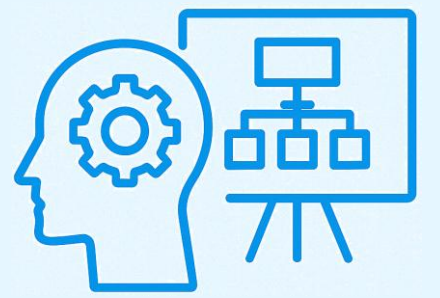
Message Broker

F5 BIG-IP LTM-4200v

DB2

Prescribed

The POM transforms the logical system architecture into a set of product and technology specifications

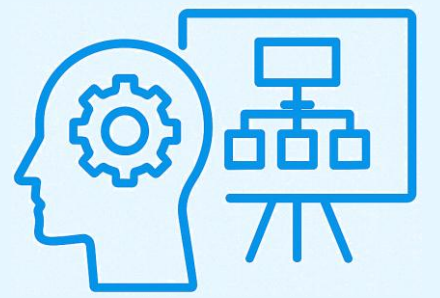


The standard technique involves four steps:

- Identifying your role in the product selection process and the definition of the selection criteria
- Performing the necessary product selections for the nodes and connections
- Defining the solution details and describing how technologies will be put together to deliver the specific SLAs
- Rationalizing the solution architecture to get to a sized POM

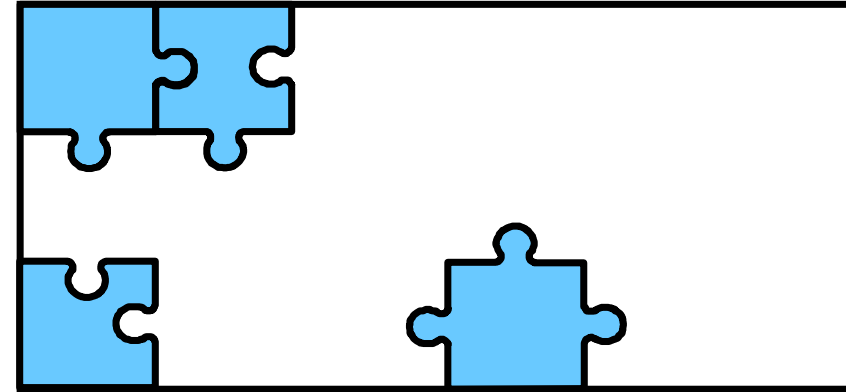


Product selection is the most important part when we build the prescribed operational model.



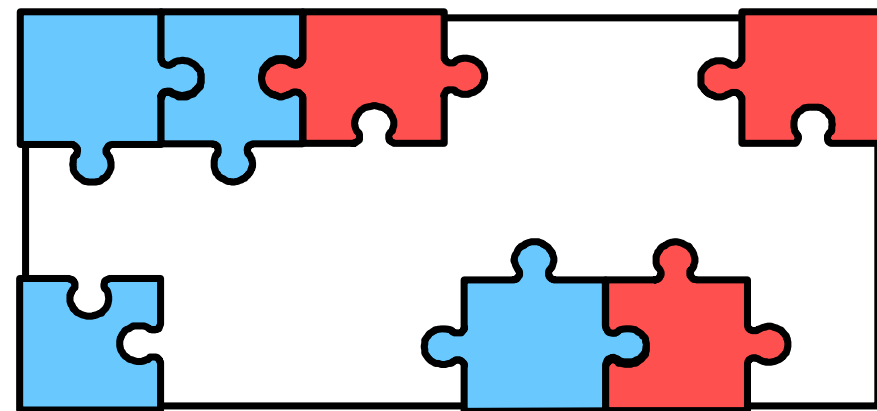
1. Identify the “Givens”

- In a client engagement
 - Usually defined in the client’s IT standards, existing Technical Architectures or current IT environment. In all cases, from the beginning you will have ensured the LOM is capable of being implemented in these technologies
- In an IBM proposal
 - IBM product, if they meet the requirements!



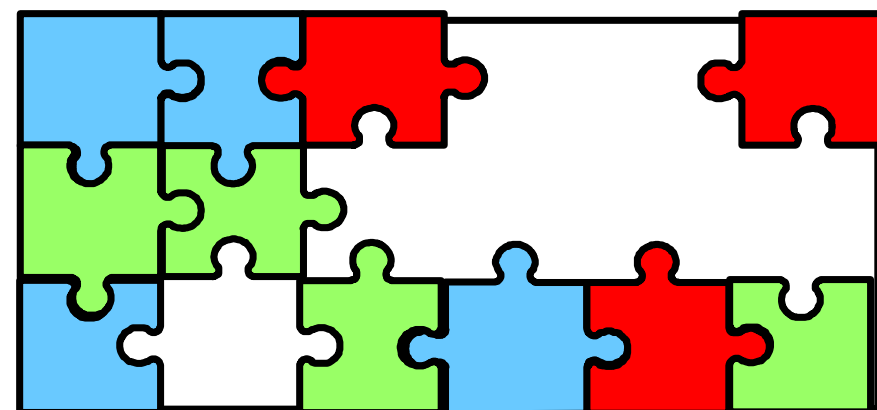
2. Spot the “Obvious” products

- “Easy to choose”, such as:
 - “Only one product/vendor meets the specs.”
 - “We all know this will be product/vendor ABC”
 - “We could do a full vendor evaluation study, but it is more efficient to choose this product/vendor now”

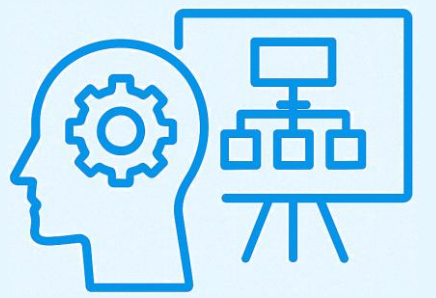


3. Look at the “Adjacent” components

- The ones that are the most difficult to choose

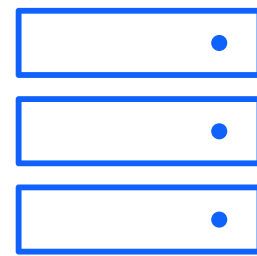


What is captured in a POM?



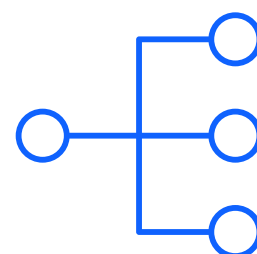
Prescribed Nodes

- Hardware technologies and products
 - Overall hardware configuration
 - Hardware specifications, such as processor speed, memory, disk configuration, etc.
 - For example, SystemP, SystemX...
- Software technologies and products
 - Software product specifications for operating system, database, middleware, directory services, security, etc., including versions
 - for example AIX Version X.x, WebSphere Application Server Liberty Profile, etc...
 - Detailed configuration of software, such as the need for multiple instances of a software product on a computer

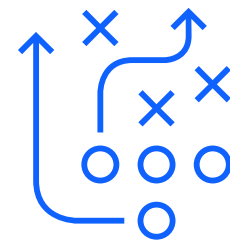
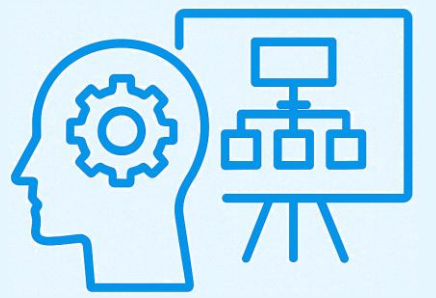


Prescribed Connections

- Technology specific, including
 - Prescribed networks, protocols, network bandwidth, latency, etc.
 - For example, 100BaseT, Fiber Optic, T1, T3, 1/10/100 Gbps Ethernet, TCP/IP, etc.



What is captured in a POM?



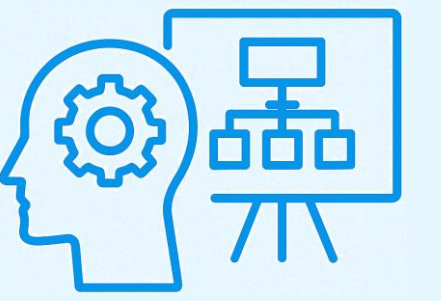
Cloud deployment model

- **Public:**
 - Shared infrastructure that lowers the cost per unit for all customers.
- **Private**
 - Provides dedicated infrastructure configured for flexibility and scalability.
 - **Managed:**
 - Takes care of infrastructure management and maintenance tasks.
 - **Hosted:**
 - Provides flexibility to self source infrastructure management and maintenance support.
- **Hybrid**
 - Mix of using shared infrastructure and dedicated infrastructure depending on NFR's.



Cloud services model

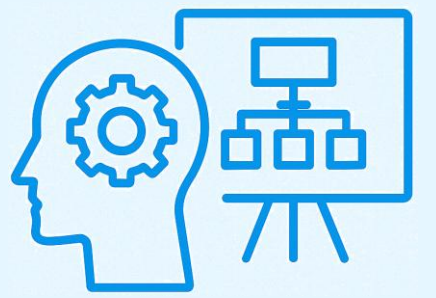
- **IaaS**
 - Infrastructure as a Service - provides raw infrastructure resources in a virtualised environment e.g. IBM Cloud.
- **PaaS**
 - Platform as a Service - provides an container/runtime environment with supporting services to build solutions e.g. IBM Cloud.
- **SaaS**
 - Software as a Service – provides application functionality as a service e.g. IBM Verse.
- **BPaaS**
 - Business Process as a Service – provides a means to increase organization efficacy by outsourcing business processes to a automated runtime e.g. through the use of IBM Blueworks Live



Traditional LOM to POM transformation



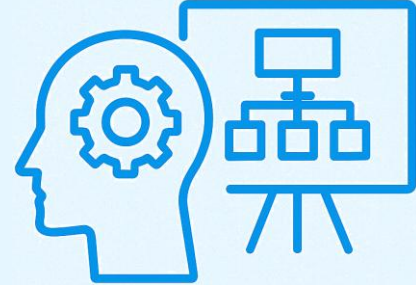
A distinction can be drawn between Prescribed Level Operational Models that emphasize



A distinction can be drawn between Prescribed Level Operational Models that emphasize:

- Component Selection (Unsize POM): Identifying the technologies and products, and the overall (qualitative) way in which they will be put together:
 - Identify the “givens”
 - Spot the “obvious” products
 - Look at the “adjacent” components
-
- Configuration (Sized POM): Defining the (quantitative) details of the solution, including the way these technologies will be put together in order to deliver the specific SLRs of the OM’s specification:
 - Performance and capacity
 - Availability
 - Security
 - Service management

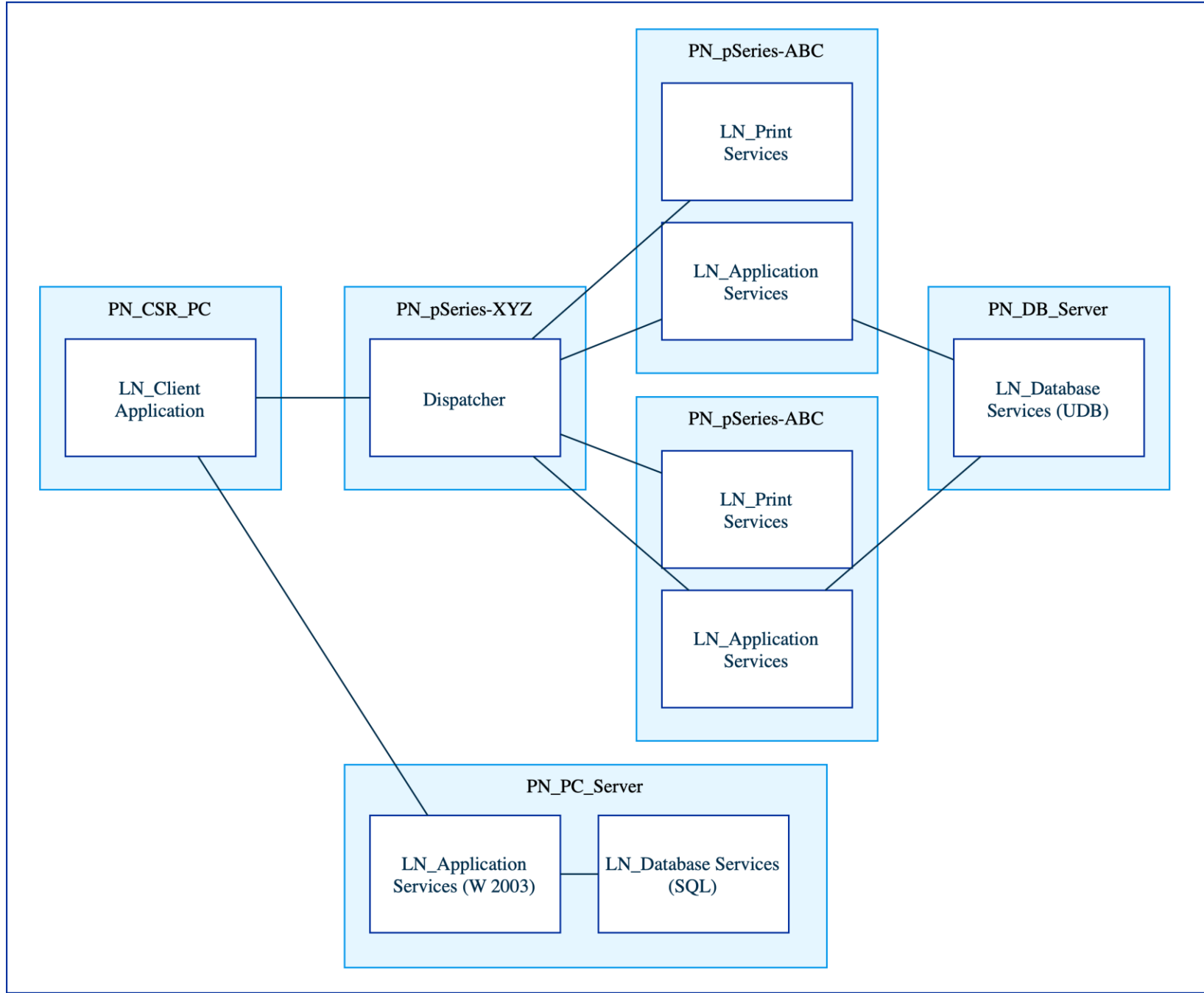
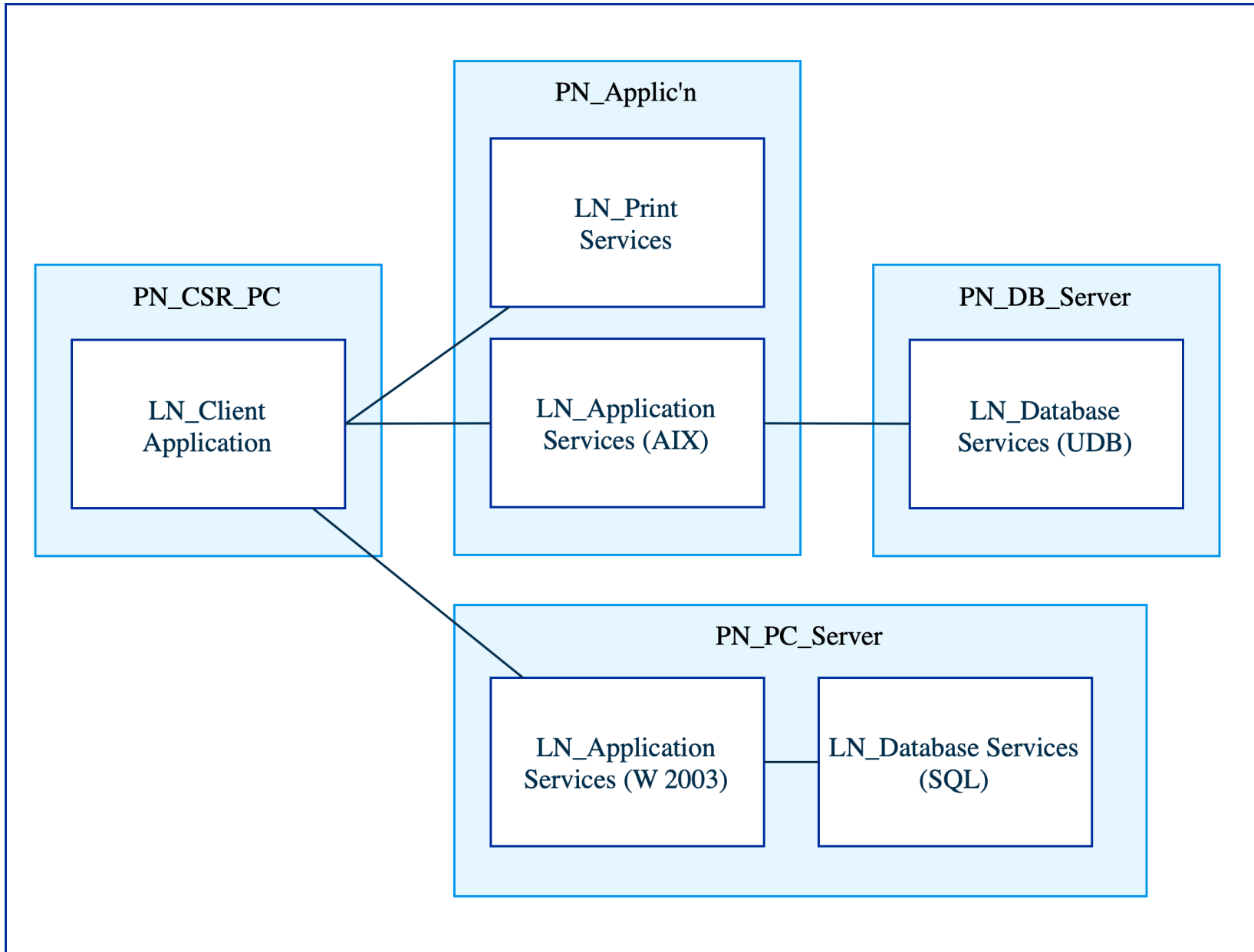
Rationalizing the design involves a process of SIZING the POM, so that performance & other characteristics can be achieved.

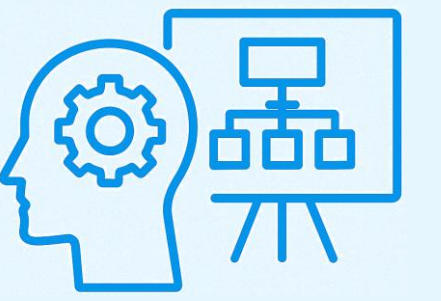


For example, the “PN_Applic’n” Prescribed node will deliver:

- The availability requirements of the Print Services, by being implemented as two Prescribed pSeries machines, with a failover capability between the “live” and “standby” machines
- The throughput requirements of Application Services, by being implemented as TWO Prescribed pSeries machines, together with a “dispatcher” Prescribed node.

This, of course, has an influence back on the POM, through the introduction of a “LN_Dispatcher” logical node

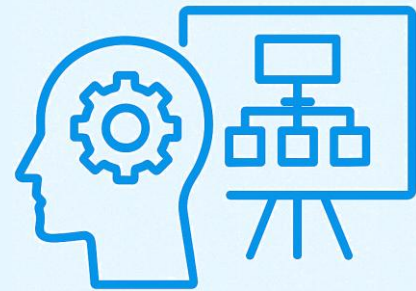




Cloud LOM to POM transformation



Cloud@POM - 4 Step approach to transforming a LOM to a cloud POM



Plan

Review the Cloud reference architectures and determine if one is a good match for the NFR requirements or tailor as required.



Define

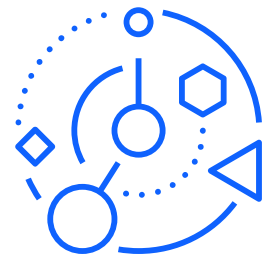
Define the type of cloud deployment model (CDM) i.e. public, private, hybrid, that meets the solution NFR's and constraints.

- Logical Service Models e.g. IaaS, PaaS, SaaS, BPaaS
- Logical Services e.g. Middleware, Database, Storage



Select

Select a cloud provider that meets NFR and CDM requirements with a sizing/pricing model that provides the best match for the solution.



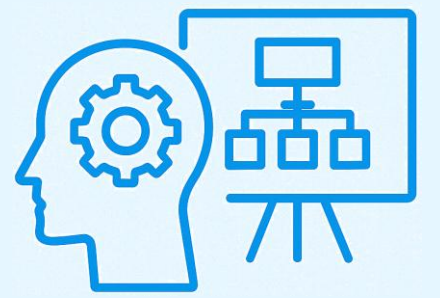
Configure

Using NFR's, LOM and cloud reference architecture as inputs

- Configure the required Prescribed cloud services offered within the chosen providers CDM.
- Configure performance, capacity, and availability by configuring Prescribed cloud services sizing/service level requirements (SLR's)
- Understand licensing model including enterprise level agreements for software/services installed within IaaS and PaaS.



Cloud@POM –Plan what kind of cloud is required for the solution



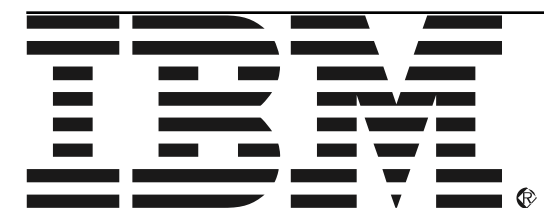
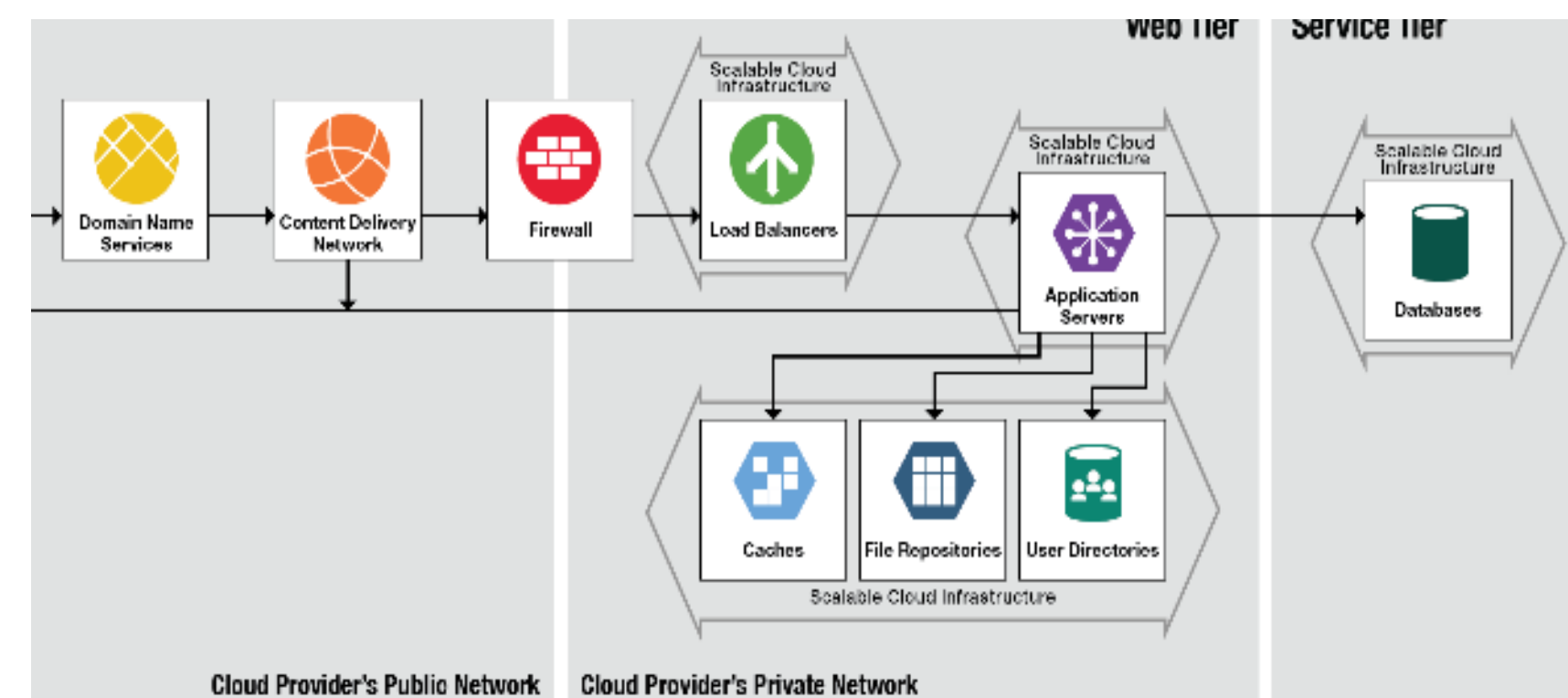
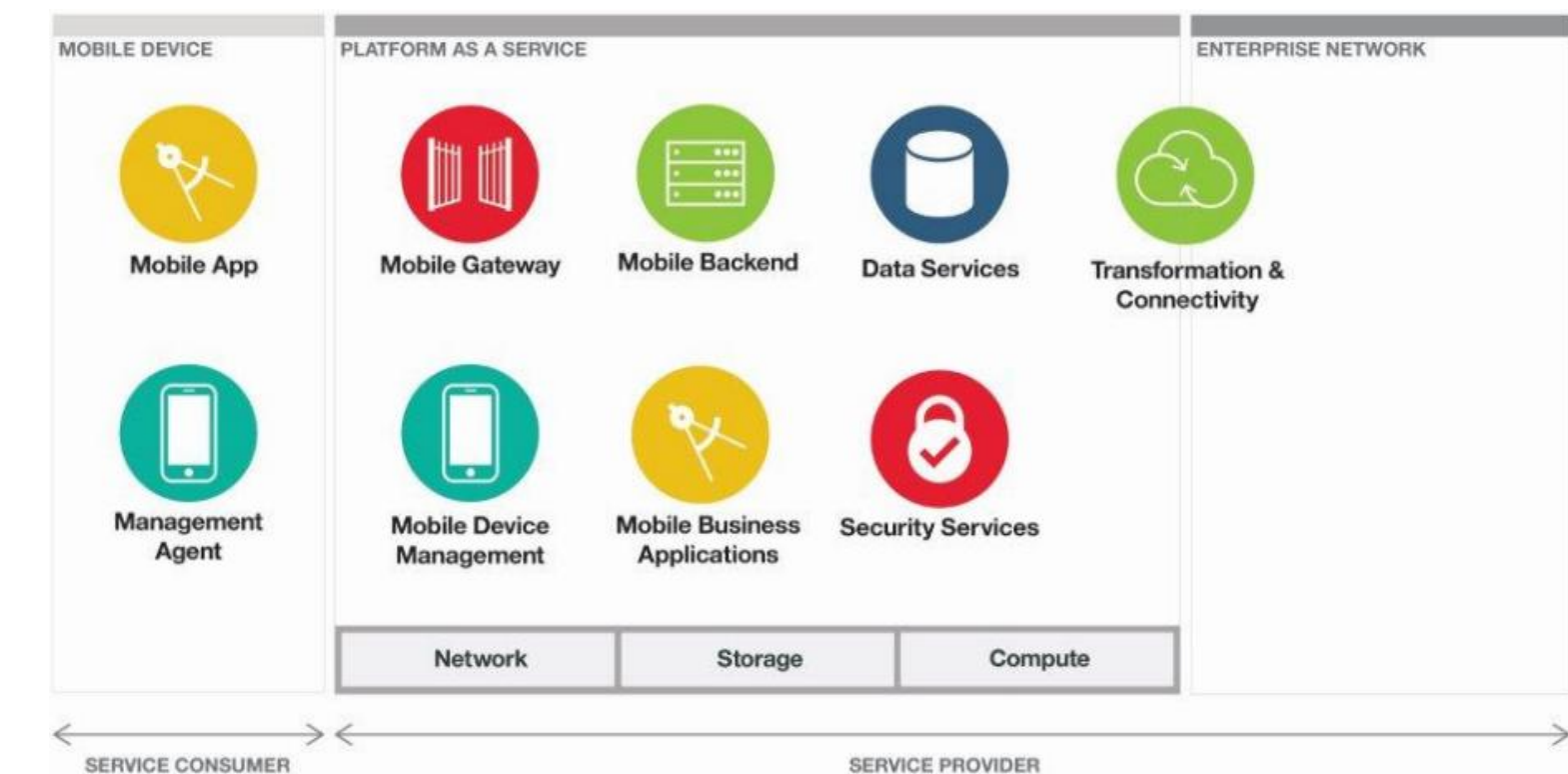
Selecting a cloud reference architecture helps leverage lessons learnt and best practices but you may still need to tailor to your needs or in the case none fit the solution define your own

Object Management Group - Cloud Standards Customer Council:

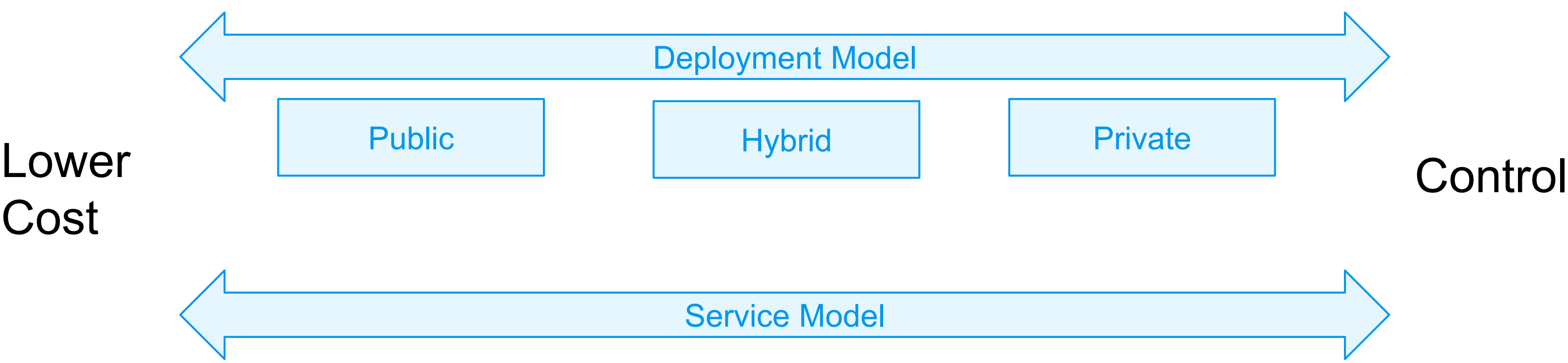
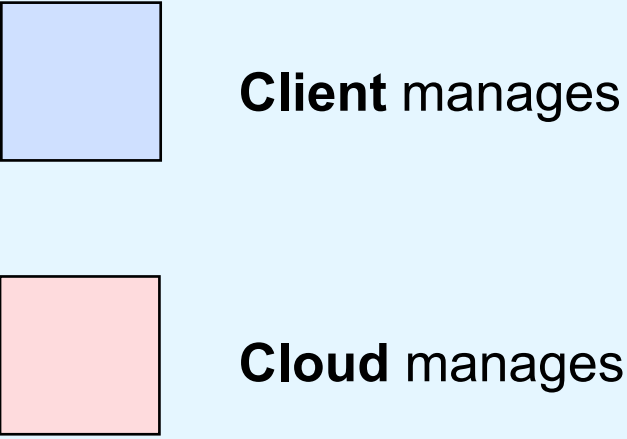
- [CLOUD CUSTOMER ARCHITECTURE FOR SECURING WORKLOADS ON CLOUD SERVICES](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR API MANAGEMENT](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR HYBRID INTEGRATION](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR ENTERPRISE SOCIAL COLLABORATION](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR E-COMMERCE](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR IOT](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR BIG DATA AND ANALYTICS V2.0](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR MOBILE](#)
- [CLOUD CUSTOMER ARCHITECTURE FOR WEB APPLICATION HOSTING V2.0](#)

IBM Hybrid Cloud Architecture Center

- <https://www.ibm.com/architectures/hybrid>

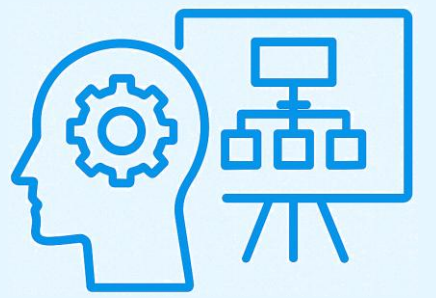


Architectural decisions need to be made to select the required deployment and service model(s) based on the NFR's and constraints while balancing value for the Customer.



SaaS	PaaS	IaaS	Traditional In-House
Business Process	Business Process	Business Process	Business Process
Applications	Applications	Applications	Applications
Data	Data	Data	Data
Runtime	Runtime	Runtime	Runtime
Middleware	Middleware	Middleware	Middleware
O/S	O/S	O/S	O/S
Virtualisation	Virtualisation	Virtualisation	Virtualisation
Servers	Servers	Servers	Servers
Storage	Storage	Storage	Storage
Networking	Networking	Networking	Networking

Cloud@POM – Selecting cloud providers



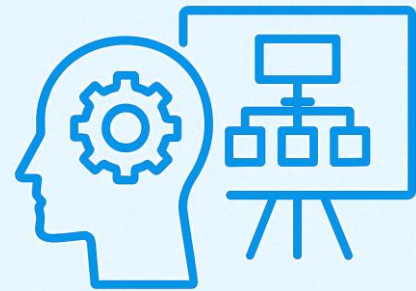
- Some considerations include:
 - Availability of the cloud platform within the client's jurisdiction.
 - Enterprise strength of the services offered by cloud providers, in particular security, availability, backup and disaster recovery.
 - The level of support offered by the cloud provider
 - Network connectivity to the cloud provider's data centre. One of the disadvantages of a cloud based solution is the potential for increased latency.
 - Content of the cloud provider's service catalogue (e.g. operating system, databases, middleware, ect).



Many others!



Cloud@POM – Configure the cloud to meet the NFR’s



- 1. Need to select a Prescribed service for each logical services identified in the LOM i.e. Database -> DashDB
- 2. Define the size of the cloud services and as you can see the major cloud providers use a T-shirt sizing model and you need to consider the NFR’s to select the “T-shirt” the best fits the solution while keeping customer value in mind

Microsoft Example

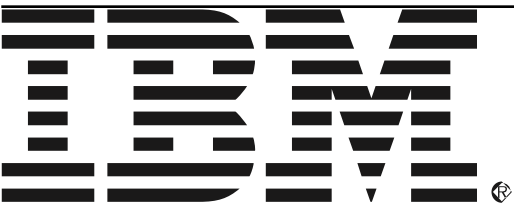
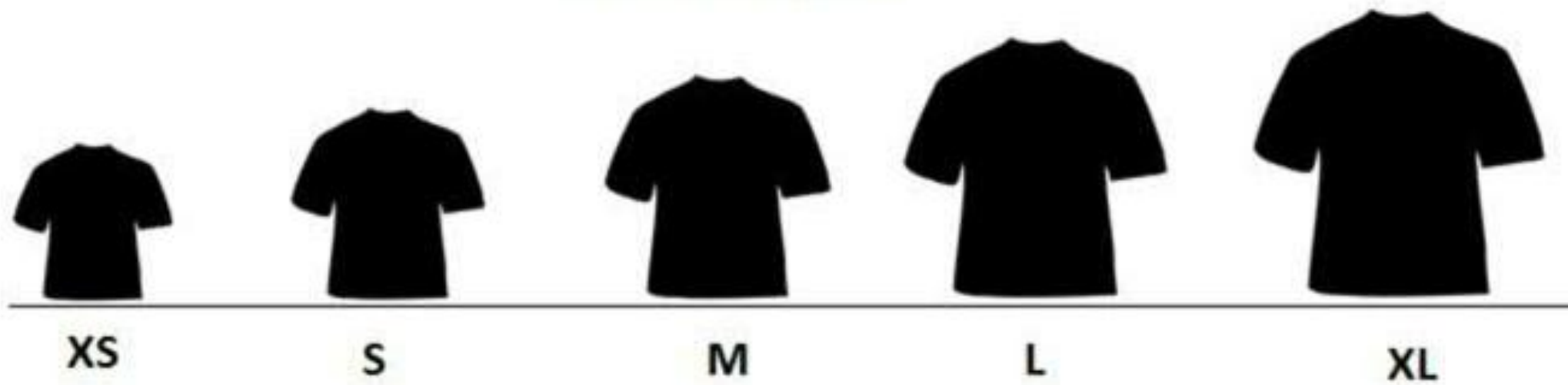
SKU Family	ACU/Core
ExtraSmall	50
Small-ExtraLarge	100
A5-7	100
Standard_A1-8v2	100
Standard_A2m-8mv2	100
A8-A11	225*
D1-14	160

IBM Example

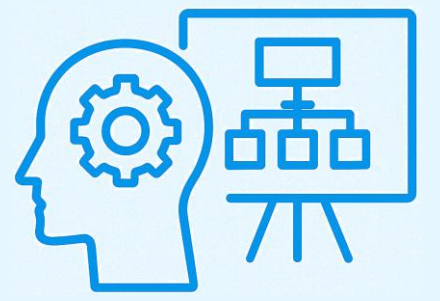
T-Shirt	vCPU	RAM (GB)	HD (GB)	Price/Hr
S	1	2	12	\$0.21
M	2	4	25	\$0.42
L	4	8	50	\$0.84
XL	8	16	100	\$1.68
XXL	16	32	200	\$3.36

SAP Example

SAP HANA Sizing
Fit-For-Purpose

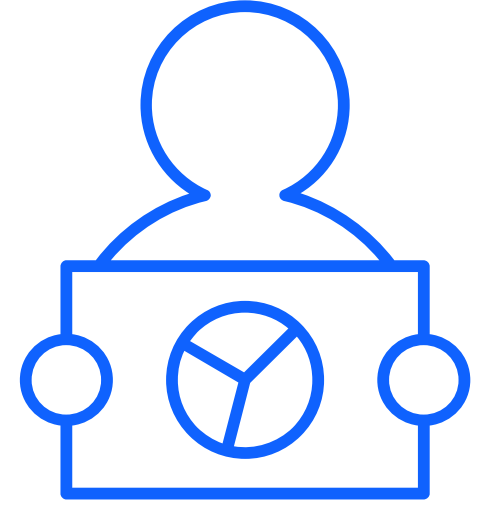
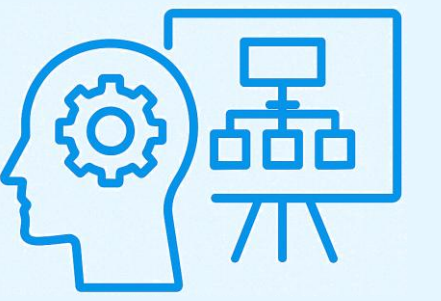


The review process leverages transaction walkthroughs to test the solution's robustness

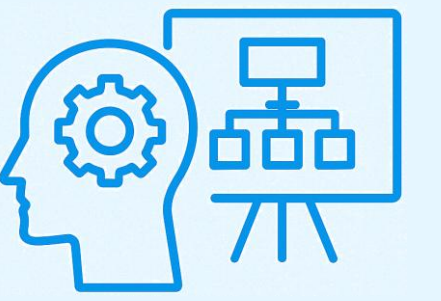


At every level of the OM, “transaction walkthroughs” using the OM walkthrough diagram are a powerful means of verifying whether the OM will work

- Analyze each transaction from start to finish following the flow of information through each Prescribed node and connection
- Visualize what would happen if there was a failure on some part of the system:
 - Would the failure be detected?
 - Would the end user be notified of the failure?
 - How will the transactions and data be recovered?
 - Who will be responsible for repair of the component?
- Analyze how the failure would impact Service Level Agreements



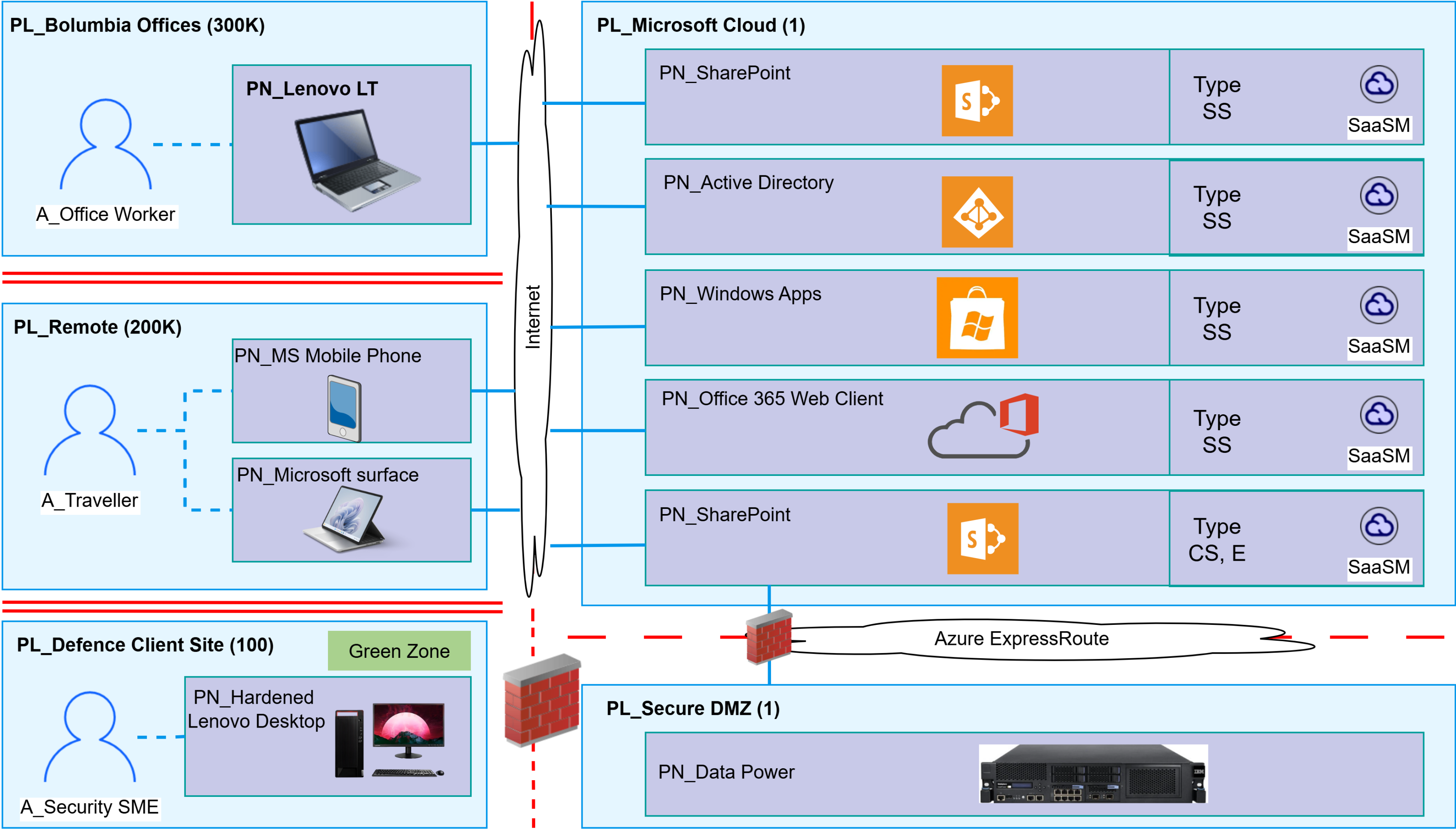
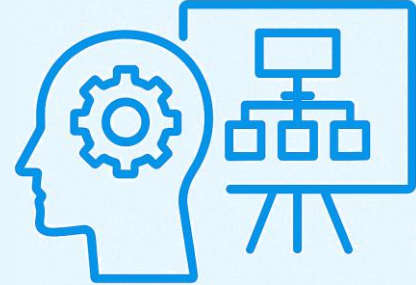
- Introduction and objectives
- What is Operational Modeling (OM) and why do we need it?
- How does OM evolve and how is it represented?
- ➔ **Constructing an Operational Model**
- Summary and references

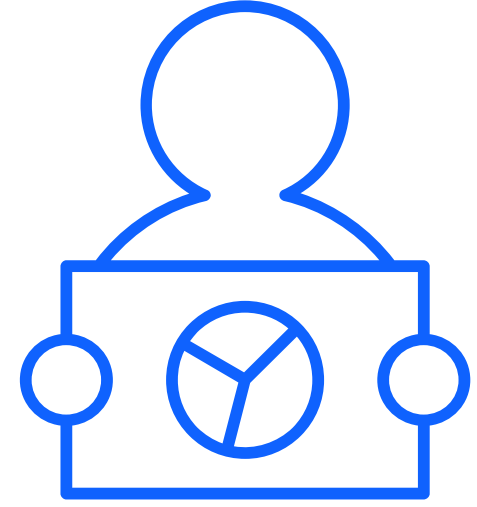
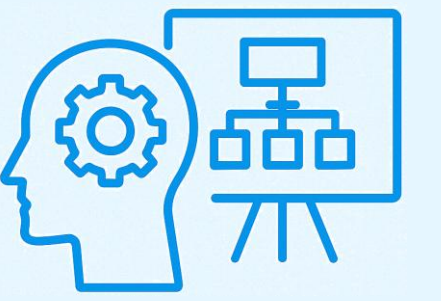


Lets transform our LOM to POM

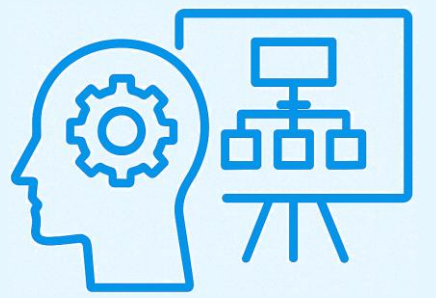


Worked Example – Word Processor Prescribed Operational Model



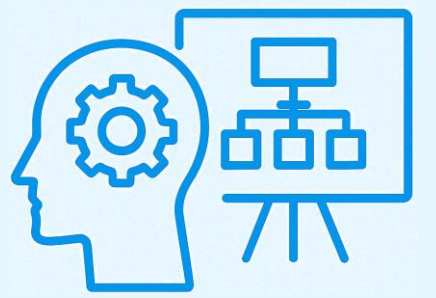


- Introduction and objectives
- What is Operational Modeling (OM) and why do we need it?
- How does OM evolve and how is it represented?
- Constructing an Operational Model
- ➔ **Summary and references**



The operational aspect of architecture:

- Focuses on the placement of the solution's DUs across the organization to achieve the business requirements of performance, availability, and other nonfunctional requirements
- Is addressed through the creation of the Operational Model at 2 different viewpoints:
 - The Logical Operational Model, which focuses on the distribution style of the application and the technical components required to support the application components
 - The Prescribed Operational Model, which details which combination of products, hardware and or cloud based services would deliver on the specification defined through the LOM



Enterprise Solutions Structure – ESS

http://ieeexplore.ieee.org/xpl/freeabs_all.jsp?arnumber=5387080

Cloud Computing

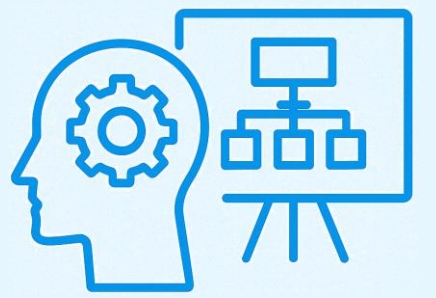
<https://www.ibm.com/developerworks/learn/cloud/index.html>

Cloud council

<http://www.cloud-council.org/>

Object Management Group (OMG) – Cloud Reference Architectures

<https://www.omg.org/cloud/deliverables/>



A number of supporting documents can be found on MethodWeb including:

- Guidance / Technique Papers
 - Developing the logical application operational model
 - Developing the logical operational model
 - Developing the Prescribed operational model
 - Designing for Availability
 - Designing for Performance and Capacity
 - Designing for Security
 - Designing for Systems Management
- Architecture Domain and related Work Product (WP) Descriptions
- Other background papers on w3 from various sources

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